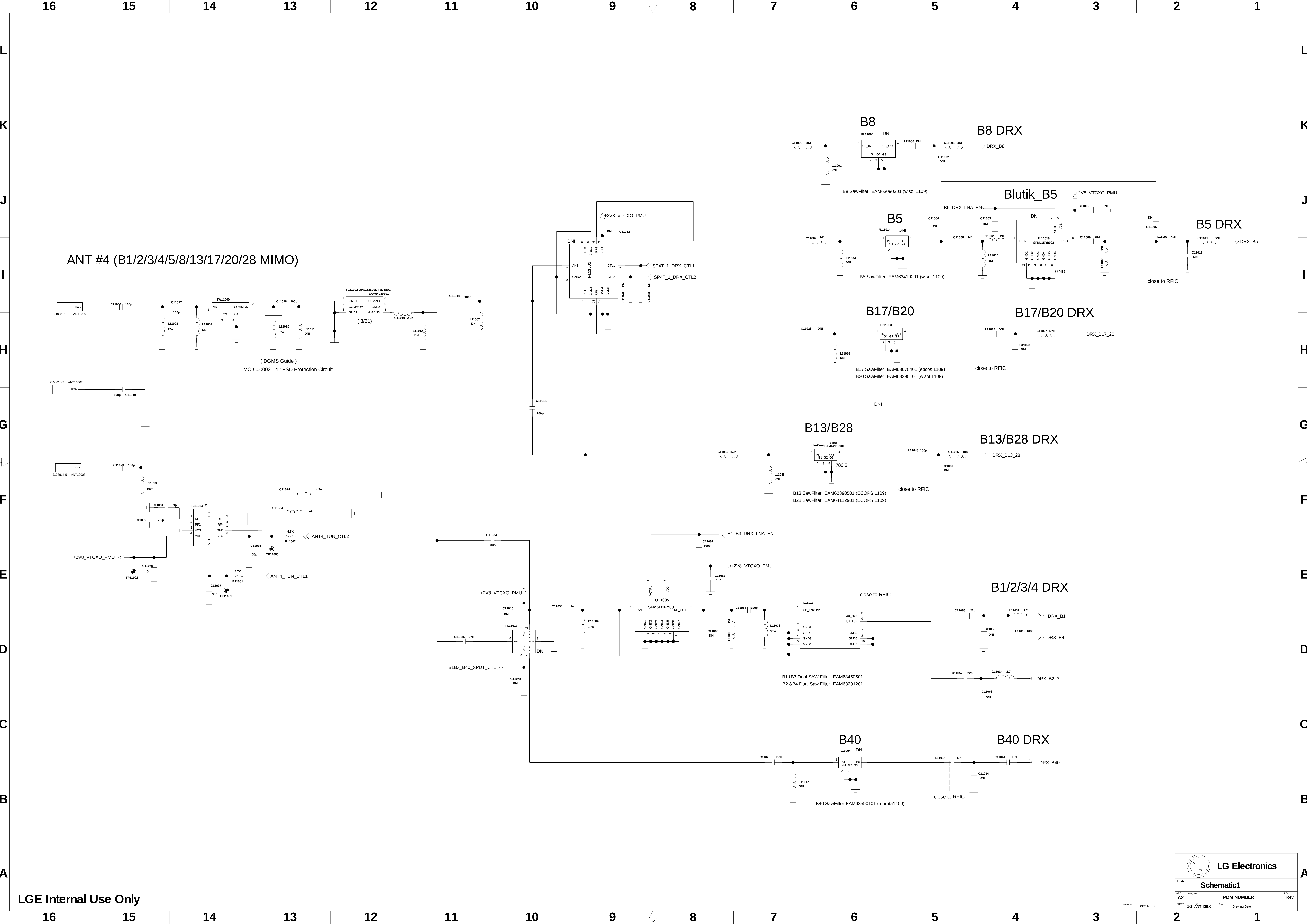
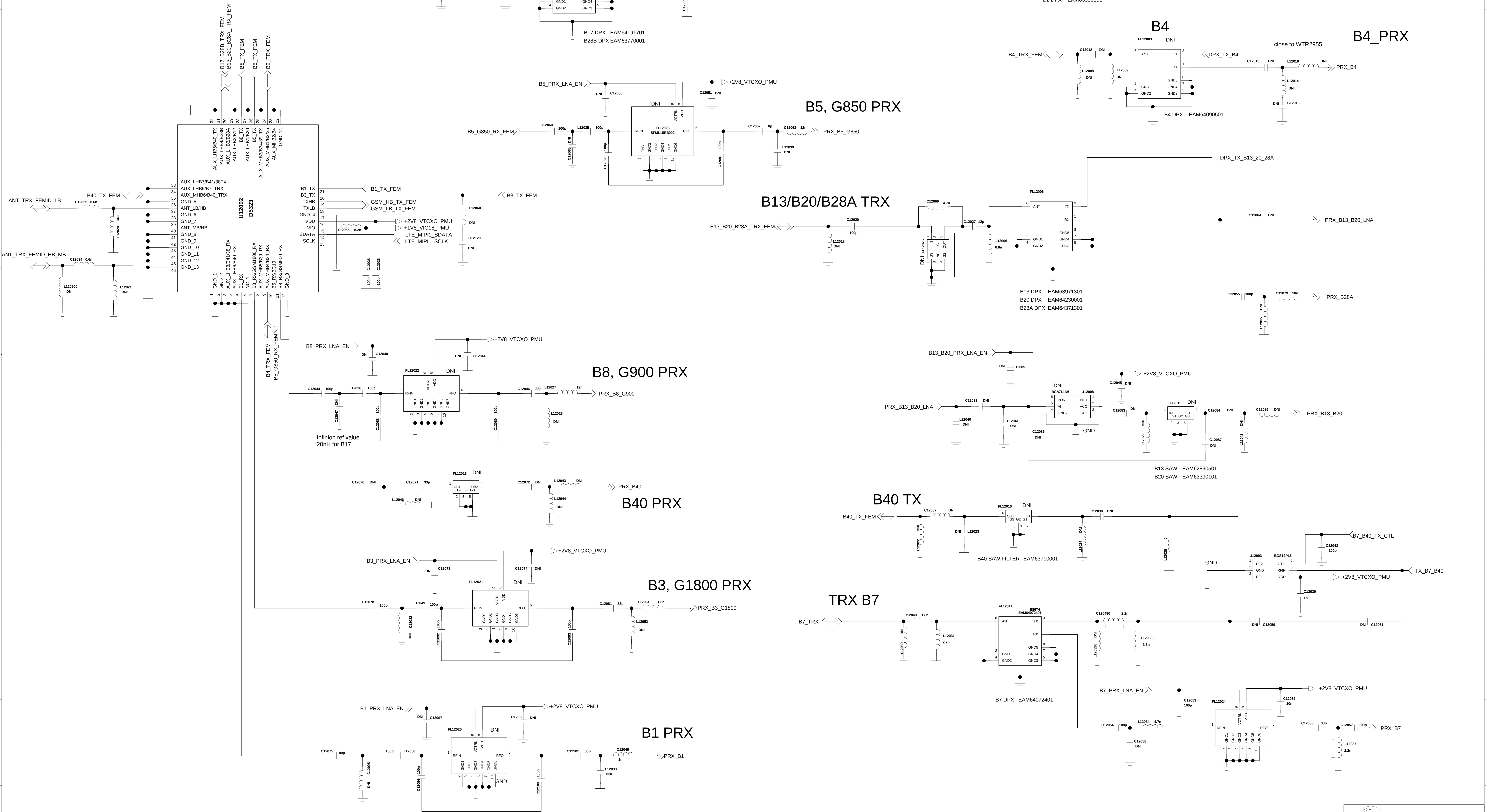




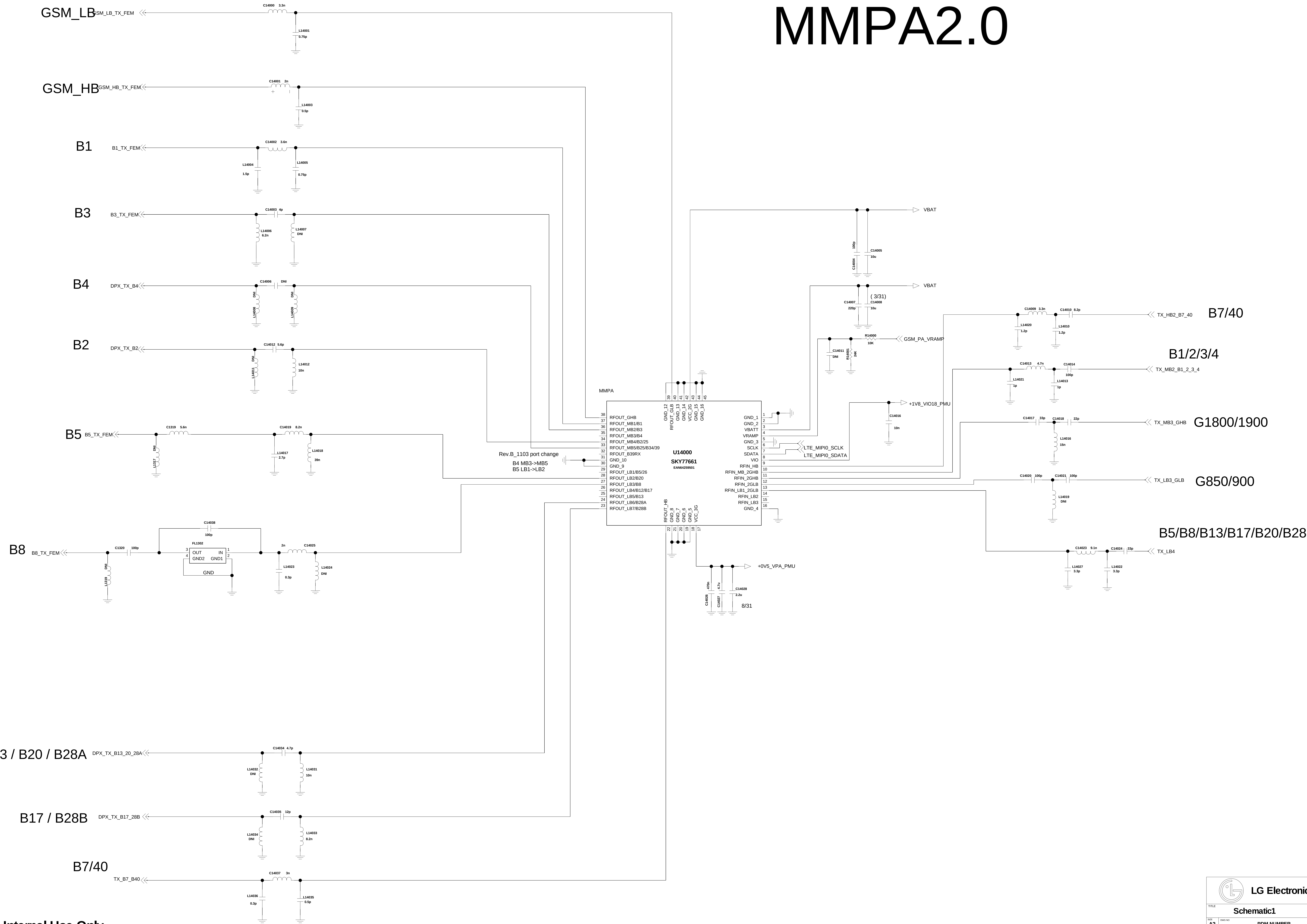
uFEMID

이더넷 SAW Front End Module(D5322) EAT63414801
B4/5/8 duplexer
G1800,TX2G LPF

Global이더넷 SAW Front End Module(D5323)EAT63375201
B1/3/5/8 duplexer



MMPA2.0

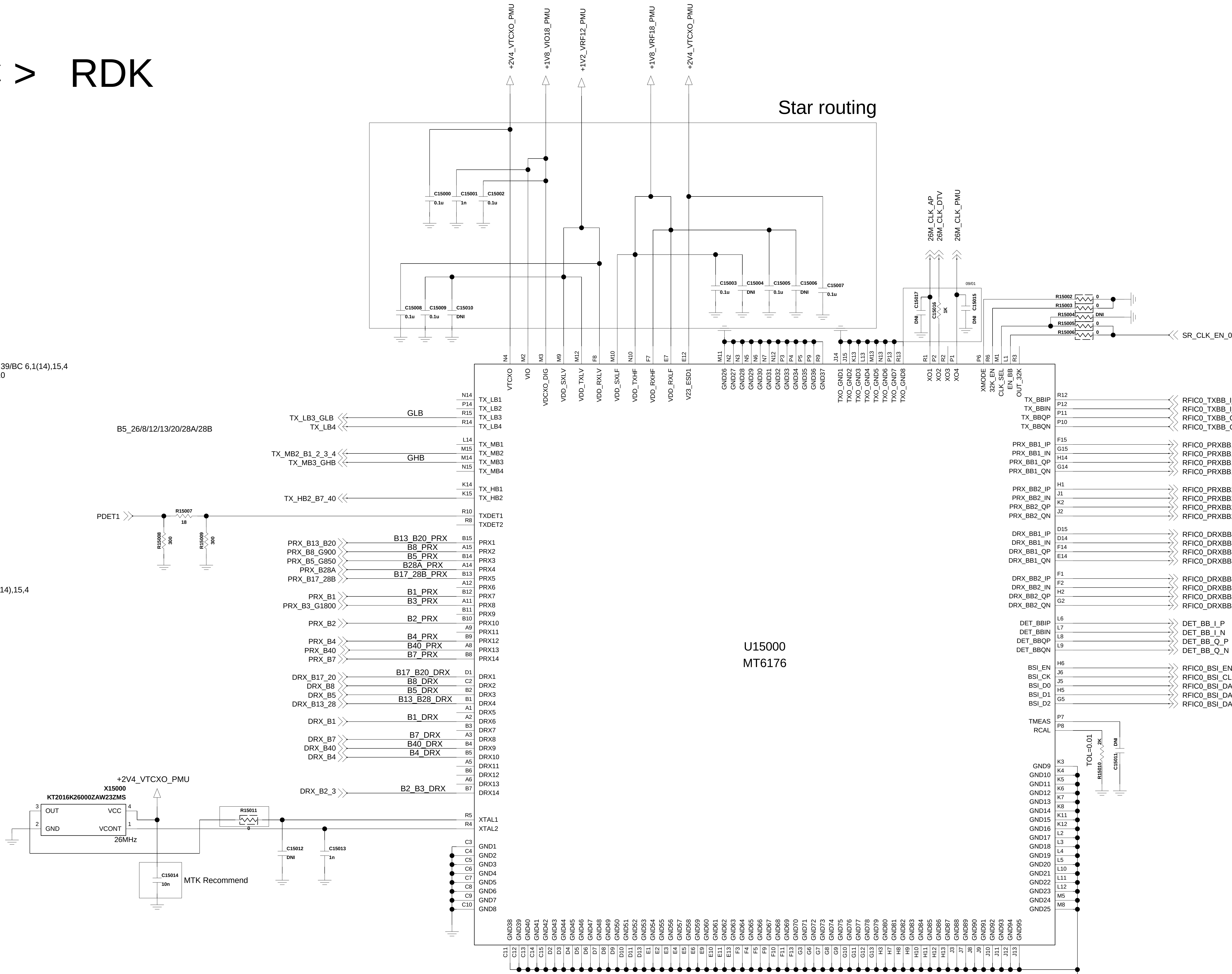


< 1-X MT6176 RFIC > RDK

H1,H2 : 7,30,38,40,41
M1,M2,M3,M4 : 1,2,3,4,9,10,11,21,23,24,25,33,34,35,36,37,39/BC 6,1(14),15,4
L1,L2,L3,L4 : 5,6,8,12,13,14,17,18,19,20,16,27,28,44/BC0,10
M3,L3 : 2G : 2,3,5,8
H1,H2 : 2300MHz-2690Mhz
M1,M2,M3,M4 : 1427MHz-2025Mhz
L1,L2,L3,L4 : 699MHz-915Mhz

HB : 7,30,38,40,41
MB : 1,2,3,4,9,10,11,21,23,24,25,33,34,35,36,37,39/BC 6,1(14),15,4
LB : 5,6,8,12,13,14,17,18,19,20,16,27,28,44/BC0,10

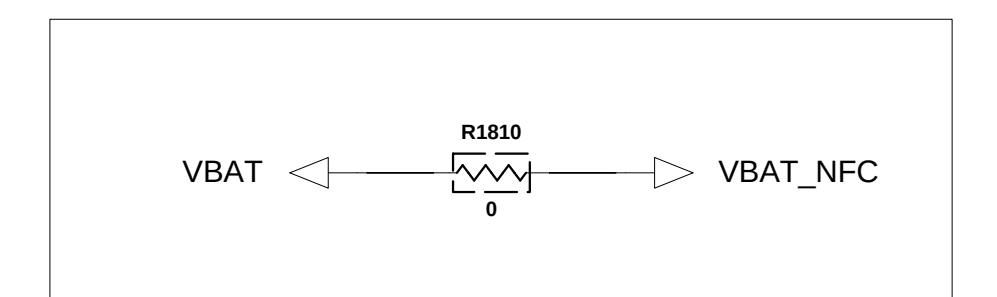
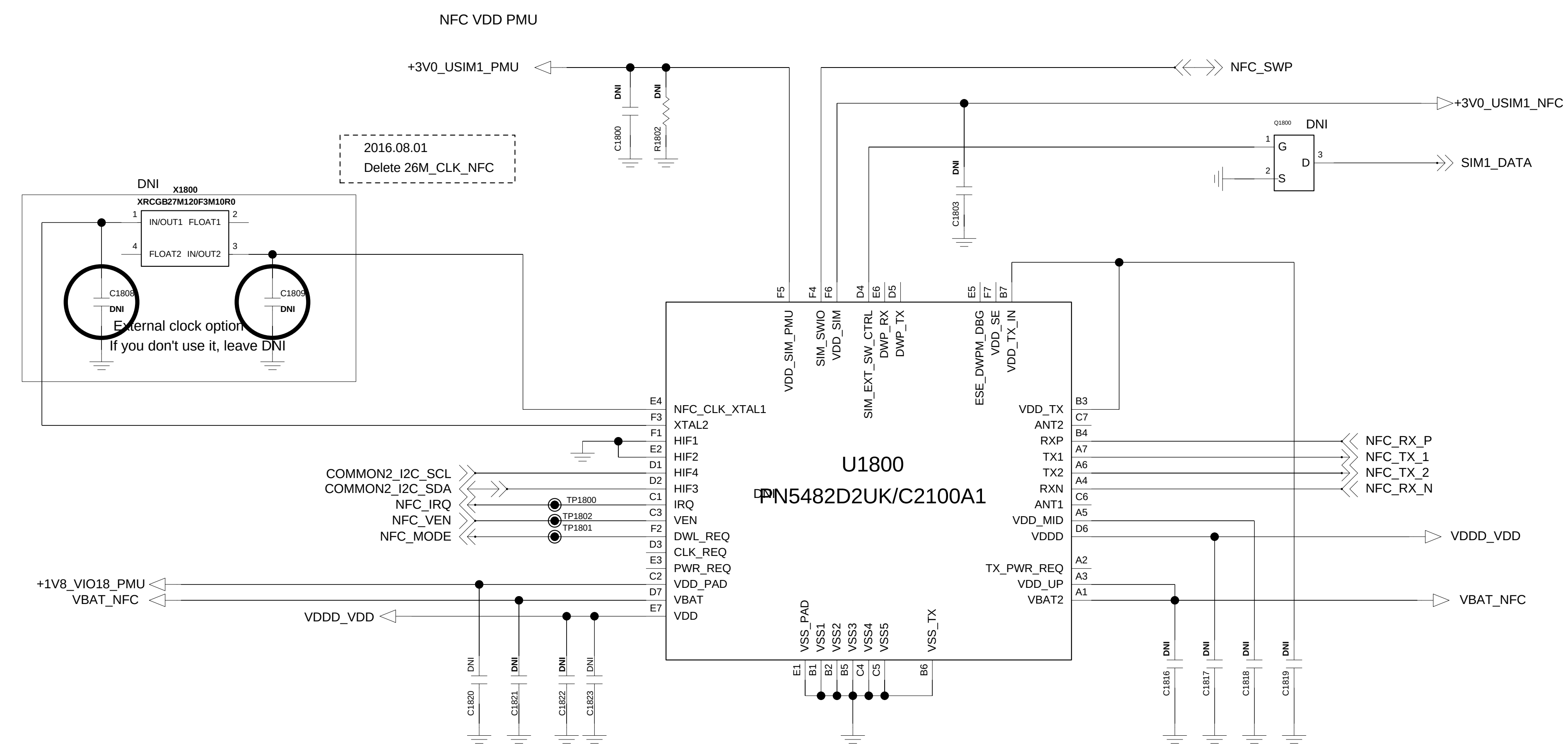
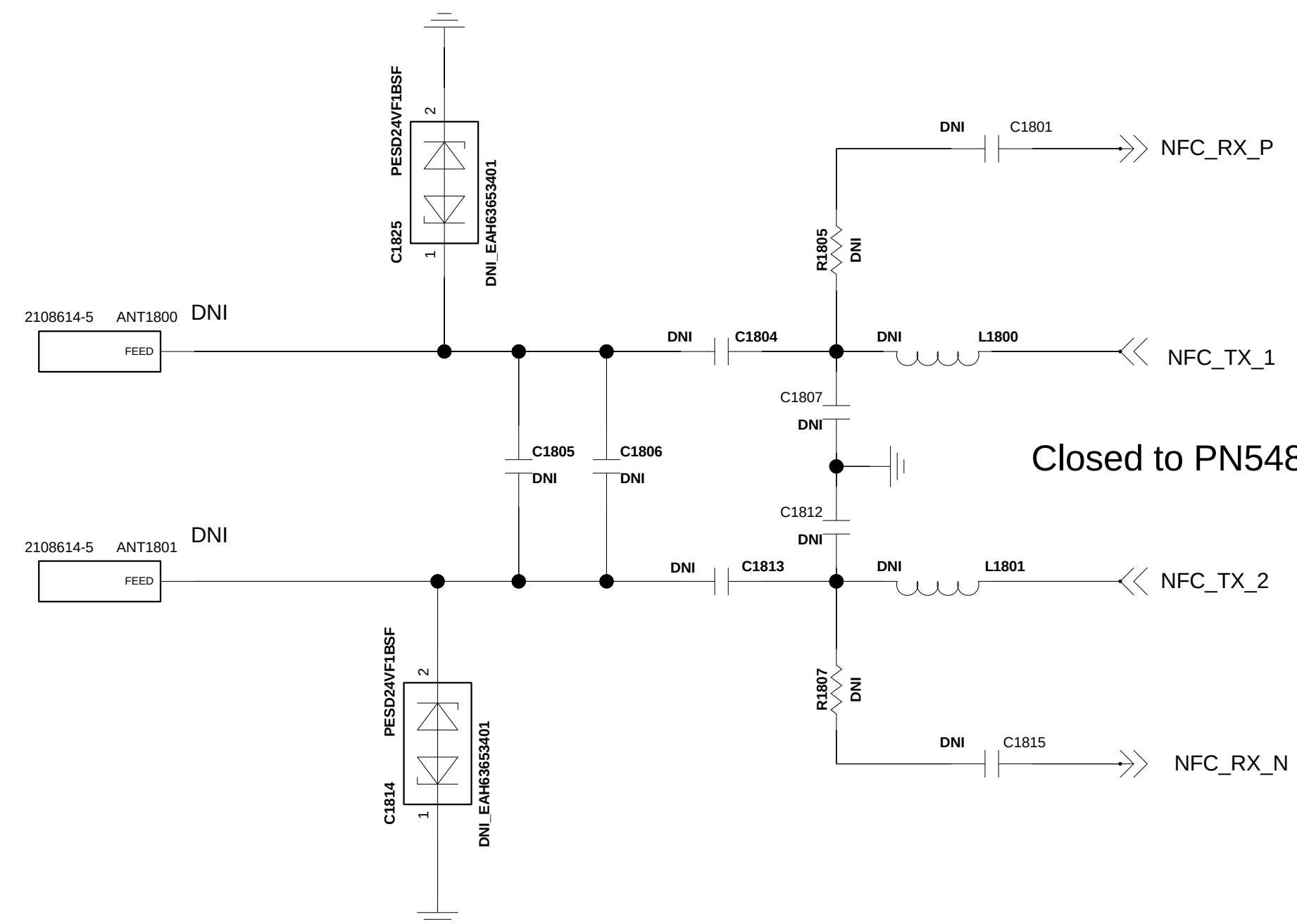
PRX14,13,12 : HB,MB 1475MHz-2690Mhz
PRX11,10,9,8 : HB,MB 1475MHz-2690Mhz
PRX7,6,5,4 : MB,LB 709MHz-2170Mhz
PRX3,2,1 : LB 703MHz-960Mhz



NFC

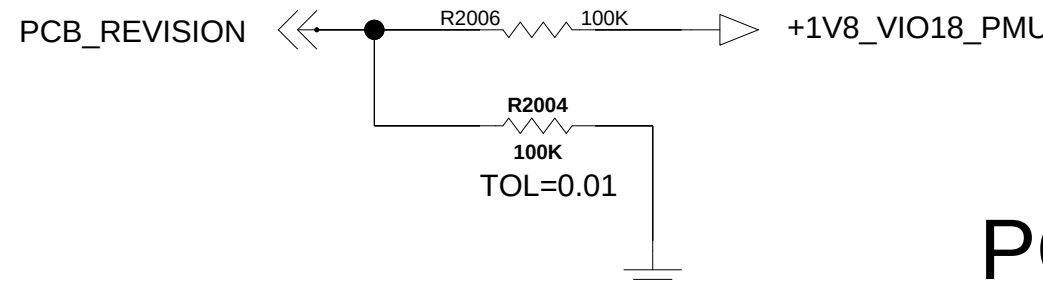
PN548_main

NFC Antenna



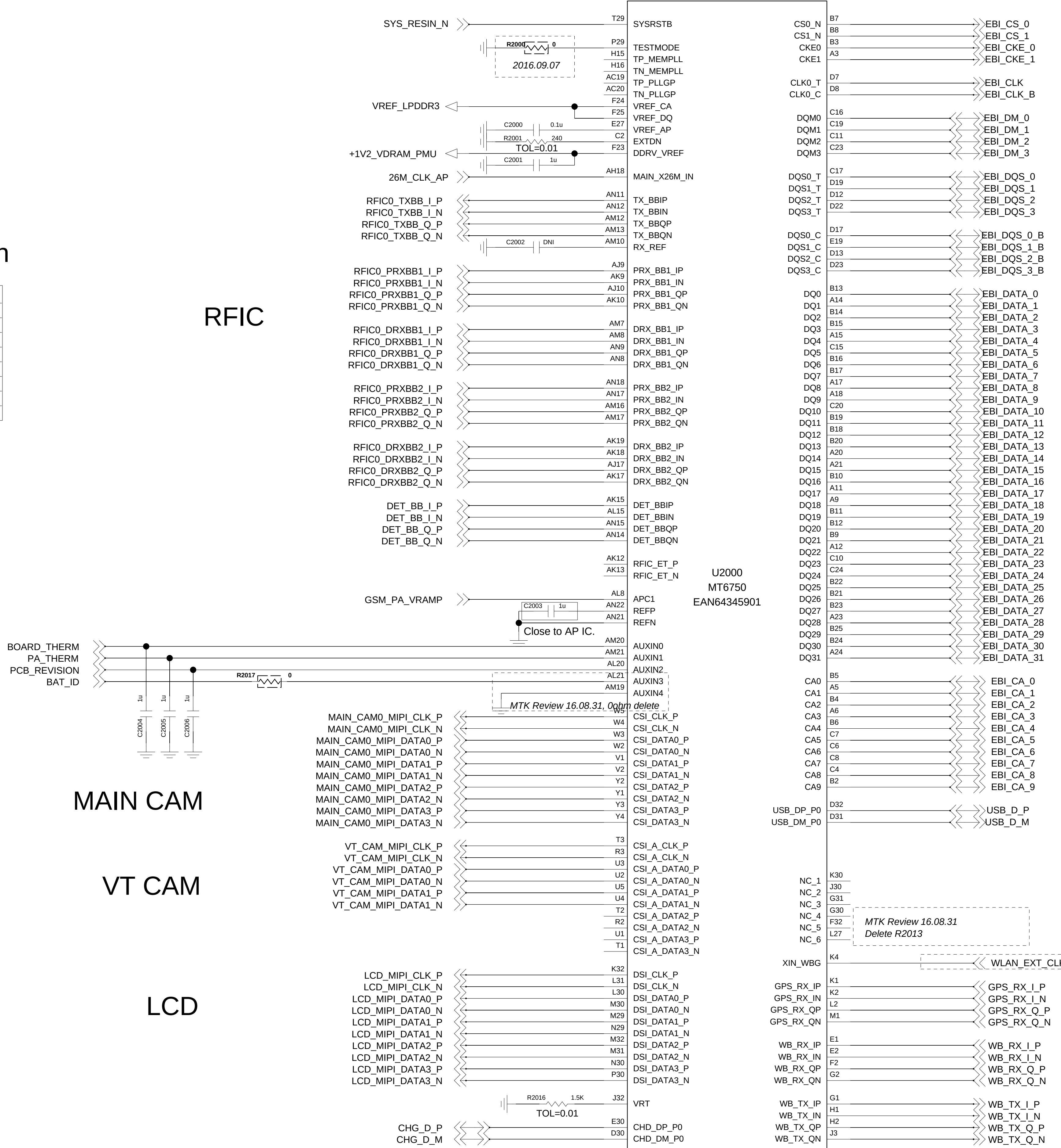
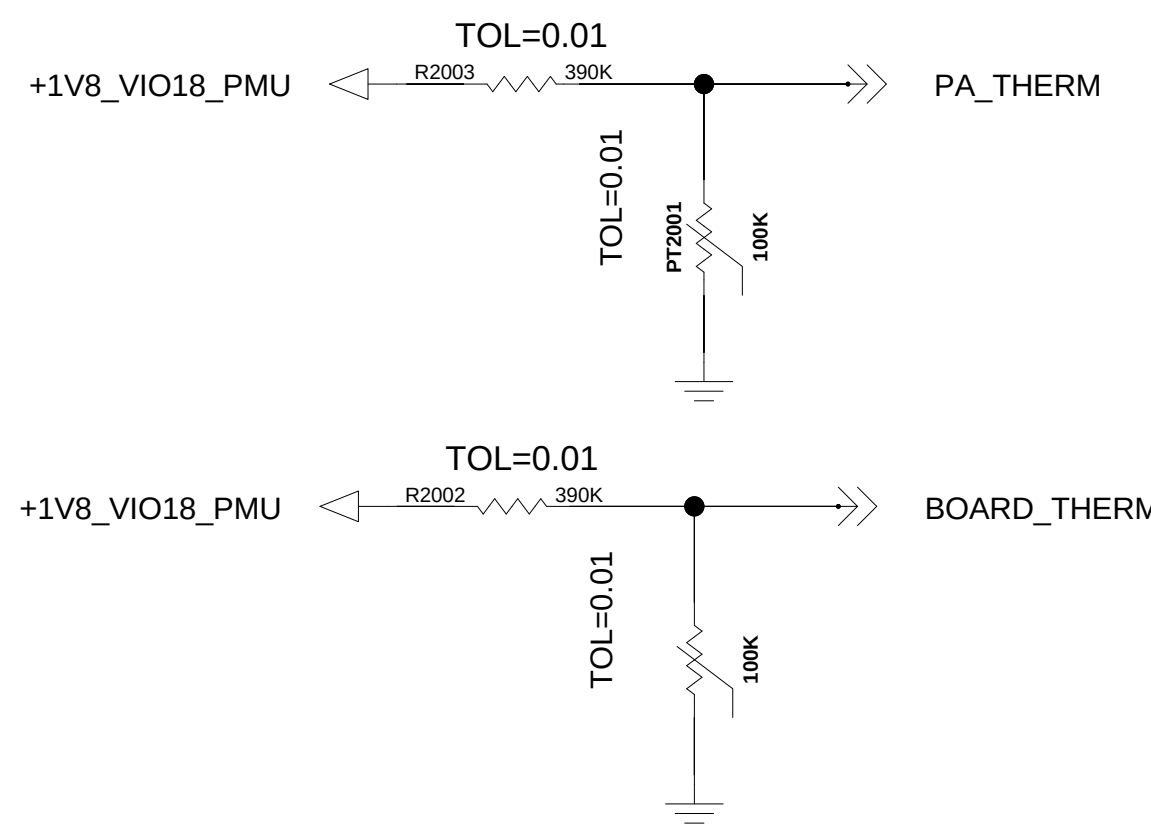
< 2-1_MT6750_DATA >

PCB Rev 1.0



PCB_Revision

Q	100K	20K	0.300
Q_1	100K	27K	0.383
A	100K	39K	0.505
B	100K	51K	0.608
C	100K	75K	0.771
1.0	100K	100K	0.900
1.1	100K	130K	1.017



MCP(LPDDR)

USB D

Default resistor of "SSUSB_VRT" can be NC if internal USB VRT is applied.

CONNECTIVITY

< 2-1_MT6750_GPIO >

MTK Review
Delete EINT_RAMDUMP 16.08.01
Delete EINT_CHARGER2_INT 16.08.31

RAM Dump Key Function possible that ther is DRAM latch Issue

SDCARD

U2000
MT6750
EAN64345901

[JTAG]

Sync. LV5

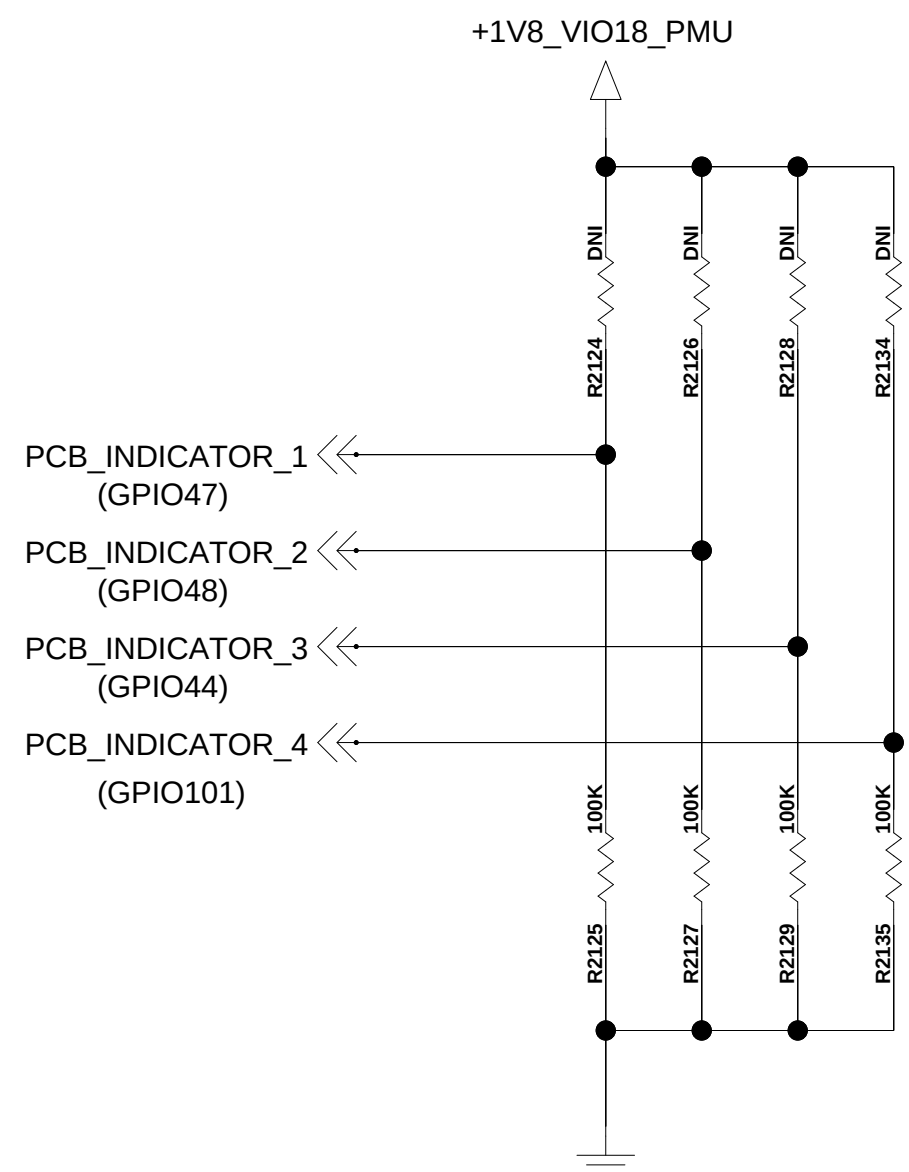
JTAG Option

GPIO153(PMI_SPI_CS)	GPIO150(Audio_DATA_MOSI)	
HI	LOW	NA
HI	HI	SPI_CSB/SPI_CLK/SPI_MO/SPI_MI
LOW	LOW	CAM_CLK0/CAM_RST0/CAM_RST1/CAM_PDN0
LOW	HI	MSDC1_CLK/MSDC_CMD/MSDC1_DAT0/MSDC1_DAT1

LV7 JTAG Option

NOTE > PWRAP_SPIO_CSN and AUD_DAT_MOSI are JTAG feature in bootstrap.

PCB INDICATOR

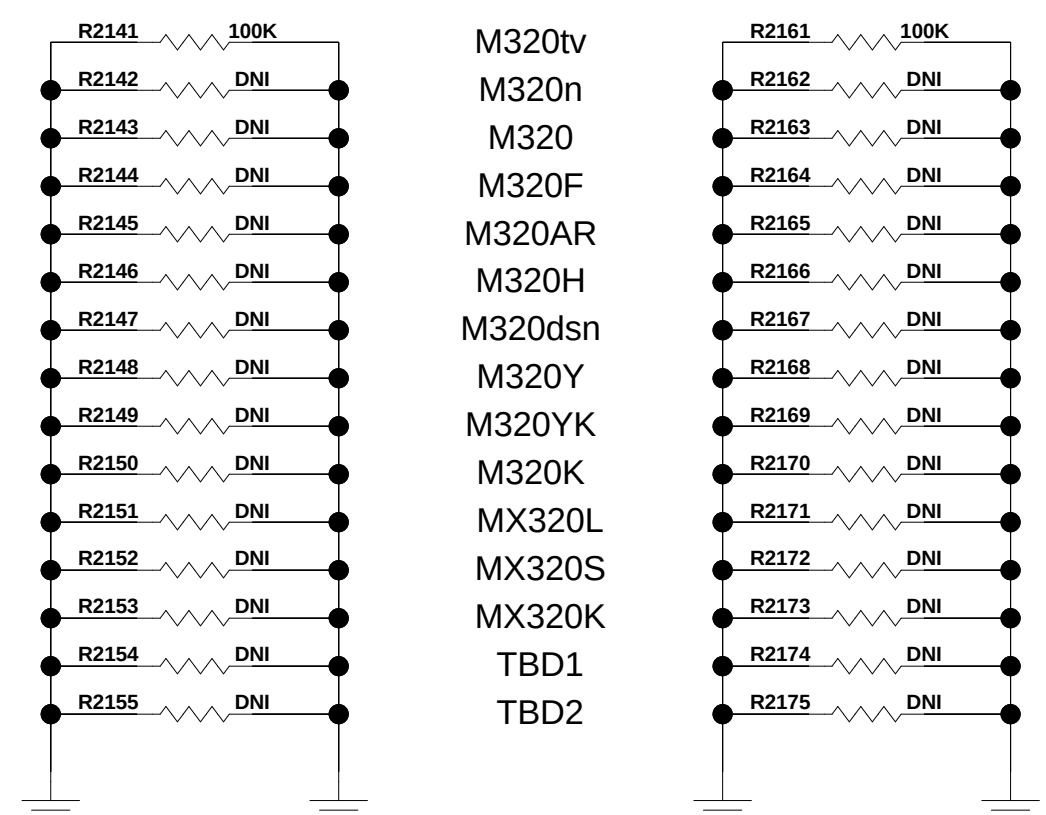


	M320tv	M320n	M320	M320F	M320AR	M320H	M320dsn	M320Y	M320YK	M320K	MX320L	MX320S	MX320K	TBD1	TBD2
PCB_INDICATOR_1	L	H	L	H	L	H	L	H	L	H	L	H	L	H	L
PCB_INDICATOR_2	L	L	H	H	L	L	H	H	L	L	H	H	L	L	H
PCB_INDICATOR_3	L	L	L	L	H	H	H	H	L	L	L	L	H	H	H
PCB_INDICATOR_4	L	L	L	L	L	L	L	L	H	H	H	H	H	H	H
	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14

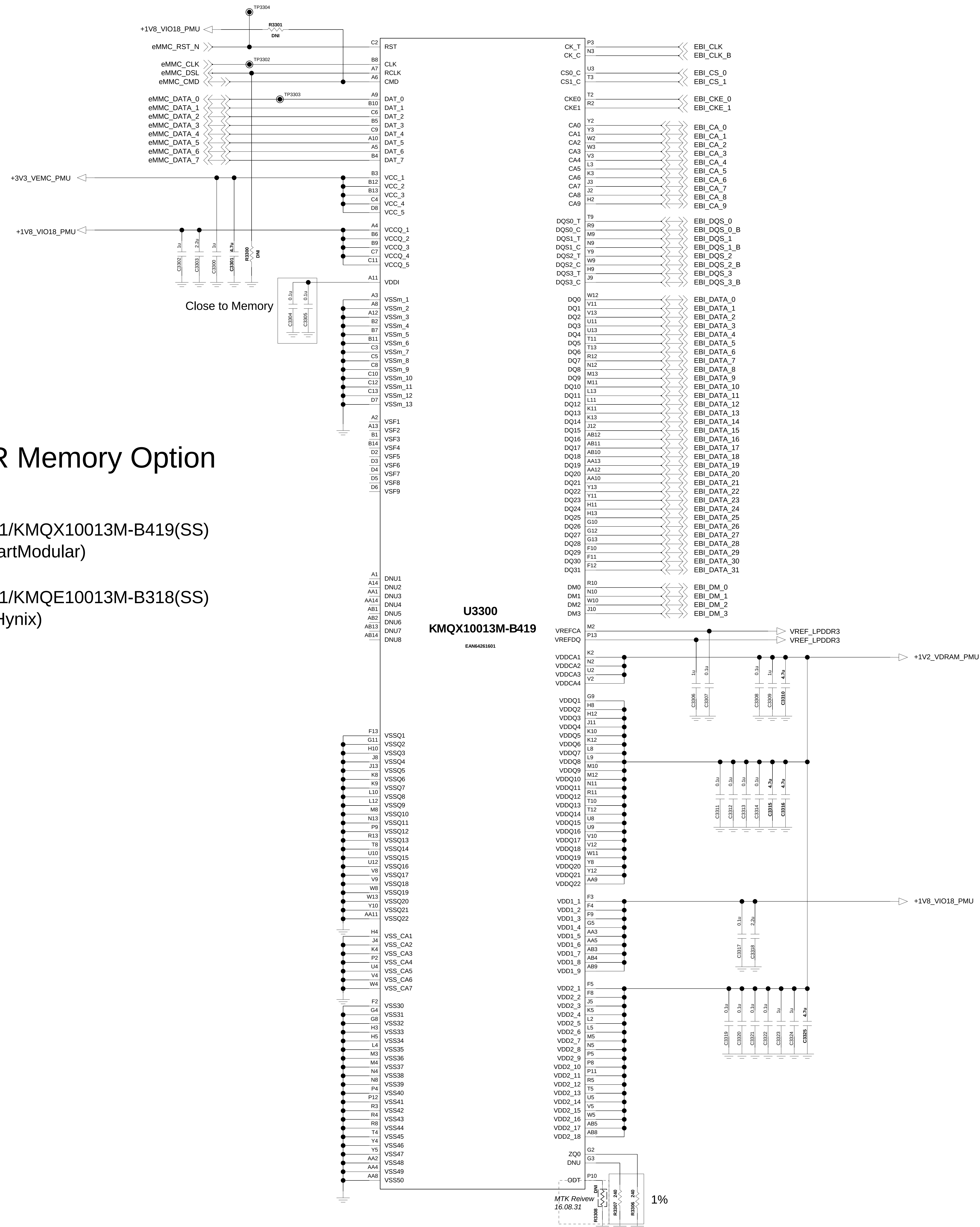
MODEL INDICATOR

TOP

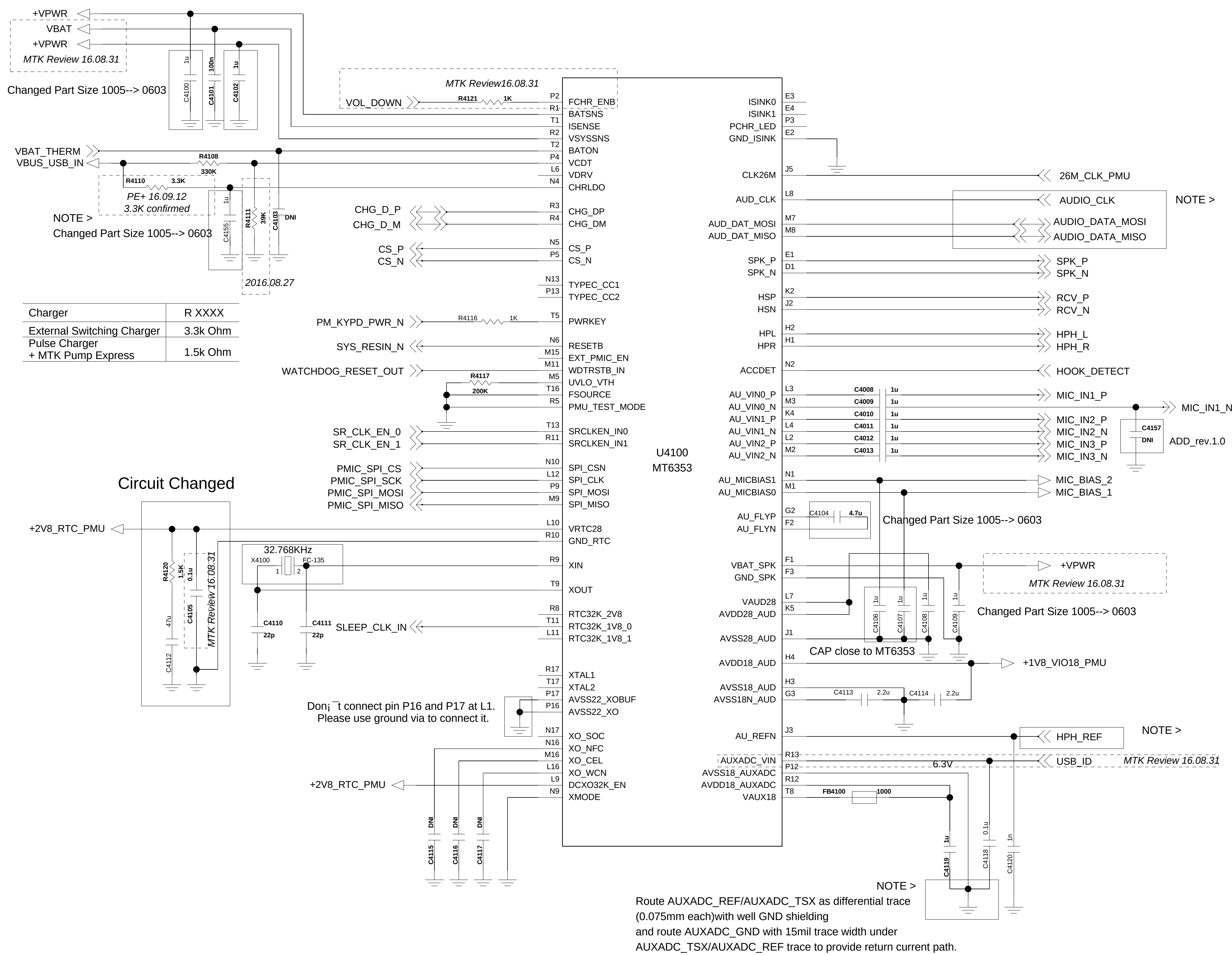
BOT



< MCP_eMMC_5_0_16Gb_DDR3_16GB_eMMC Hynix >



<4-1-15-1_PMIC_MT6353_DATA >

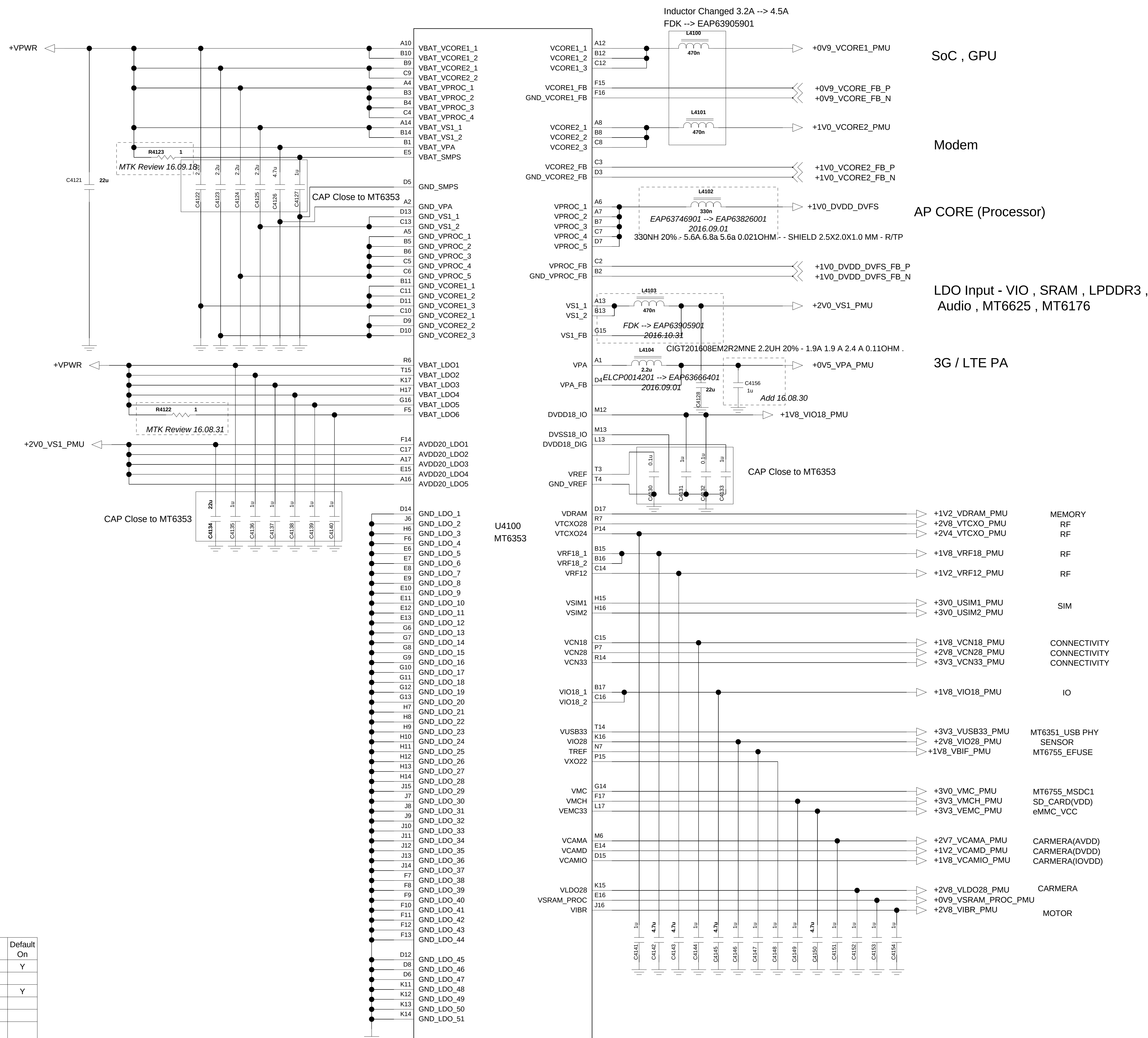


BUCK & LDO

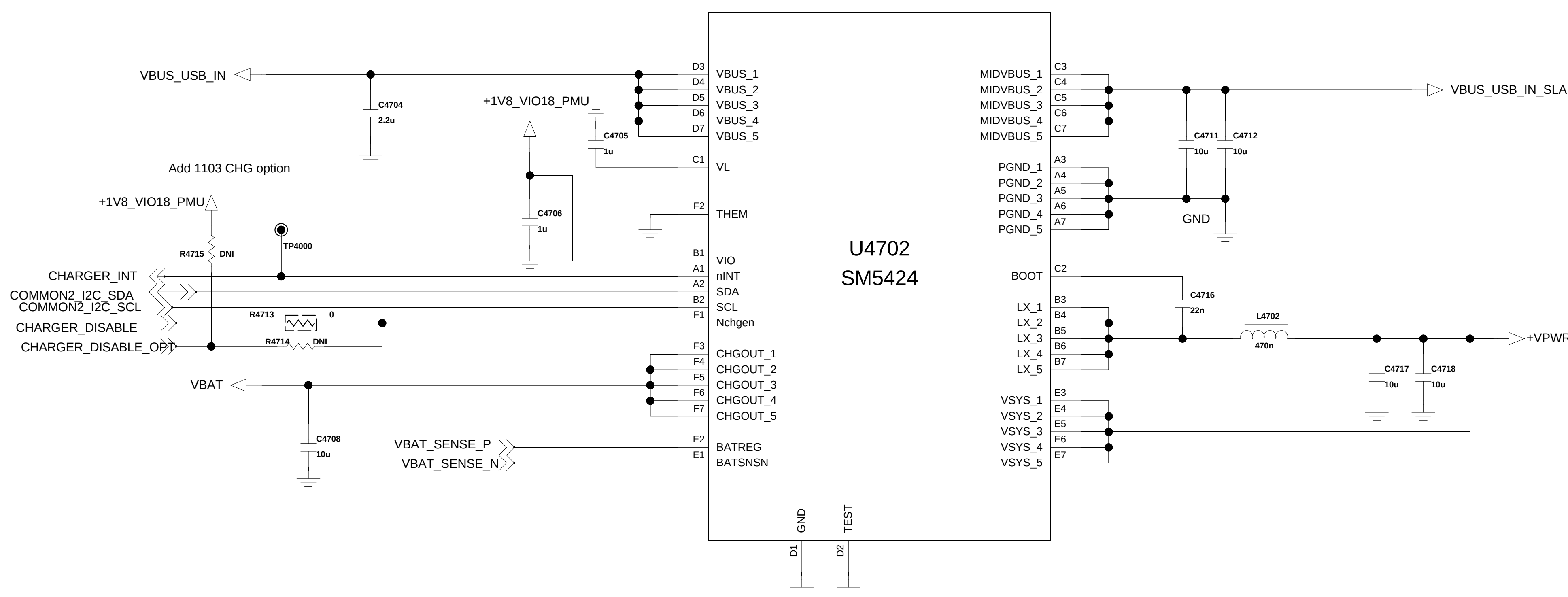
Function	Default Voltage	Vout	Voltage Step [mV]	Rated Current	Default On
VCORE1	0.9V	0.600 - 1.310 V	6.25	3,000mA	Y
VCORE2	1.0V	0.600 - 1.310 V	6.25	3,500mA	Y
VPROC	1.0V	0.600 - 1.310 V	6.25	5,000mA	Y
VS1	2.0V	1.950 - 2.200 V	12.5	1,900mA	Y
VPA	0.5V	0.500 - 3.400 V	50	800mA	N

Function	Type	Input Power Domain	Default Voltage	Vout	Rated Current	Default On
VDRAM	SLDO	AVDD20_LDO1	1.24 V	1.24 V	1,000mA	Y
VTCXO28	ALDO	VBAT_LDO51	2.8 V	2.8 V	40mA	
VTCXO24	ALDO	VBAT_LDO52	2.375 V	2.375 V	40mA	Y
VRF18	SLDO	AVDD20_LDO5	1.825 V	1.825 V	400mA	
VRF12	SLDO	AVDD20_LDO5	1.2 V	1.2 V	150mA	
VSIM1	DLDO	VBAT_LDO4	1.8 V	1.7 / 1.8 / 1.86 / 2.76 / 3.0 / 3.1 V	50mA	
VSIM2	DLDO	VBAT_LDO4	1.8 V	1.7 / 1.8 / 1.86 / 2.76 / 3.0 / 3.1 V	50mA	
VCN18	SLDO	AVDD20_LDO2	1.8 V	1.8 V	150mA	
VCN28	ALDO	VBAT_LDO1	2.8 V	2.8 V	40mA	
VCN33	DLDO	VBAT_LDO2	3.3 V	3.3 / 3.4 / 3.5 / 3.6 V	350mA	
VIO18	SLDO	AVDD20_LDO3	1.8 V	1.8 V	600mA	Y
VUSB33	DLDO	VBAT_LDO2	3.3 V	3.3 V	50mA	Y
VIO28	DLDO	VBAT_LDO4	2.8 V	2.8 V	200mA	Y
VREF	Power Switch	VBAT_LDO1	1.8 V	1.8 V	1mA	
VXO22	ALDO	VBAT_LDO52	2.2 V	2.2 V	20mA	Y
VMC	DLDO	VBAT_LDO5	3.0 V	1.8 / 2.9 / 3.0 / 3.3 V	200mA	
VMCH	DLDO	VBAT_LDO5	3.0 V	2.9 / 3.0 / 3.3 V	800mA	
VEMC33	DLDO	VBAT_LDO3	3.0 V	2.9 / 3.0 / 3.3 V	800mA	Y
VCAMA	ALDO	VBAT_LDO1	2.8 V	2.5 / 2.8 V	200mA	
VCAMD	SLDO	AVDD20_LDO2	1.3 V	0.9 / 1.0 / 1.1 / 1.22 / 1.3 / 1.5 / 1.8 V	<=1.5V : 500mA 1.8V : 300mA	
VCAMIO	SLDO	AVDD20_LDO2	1.8 V	1.8 V	200mA	
VLD028	DLDO	VBAT_LDO4	2.8 V	2.8 V	300mA	
VSRAM_PROC	SLDO	AVDD20_LDO4	1.1 V	0.6 - 1.31 V	400mA	Y
VIBR	DLDO	VBAT_LDO4	2.8 V	1.2 / 1.3 / 1.5 / 1.8 / 2.5 / 2.8 / 3.0 / 3.3 V	100mA	
VAUX18	ALDO	VBAT_LDO1	1.8 V	1.8 V	40mA	Y
VAUD28	ALDO	VBAT_LDO1	2.8 V	2.8 V	40mA	Y
VRTC28	VRTC28	VBAT_LDO1	2.8 V	2.8 V	2mA	Y
VDIG18	DVDD18_DIG	VBAT_LDO2	1.8 V	1.8 V	20mA	Y

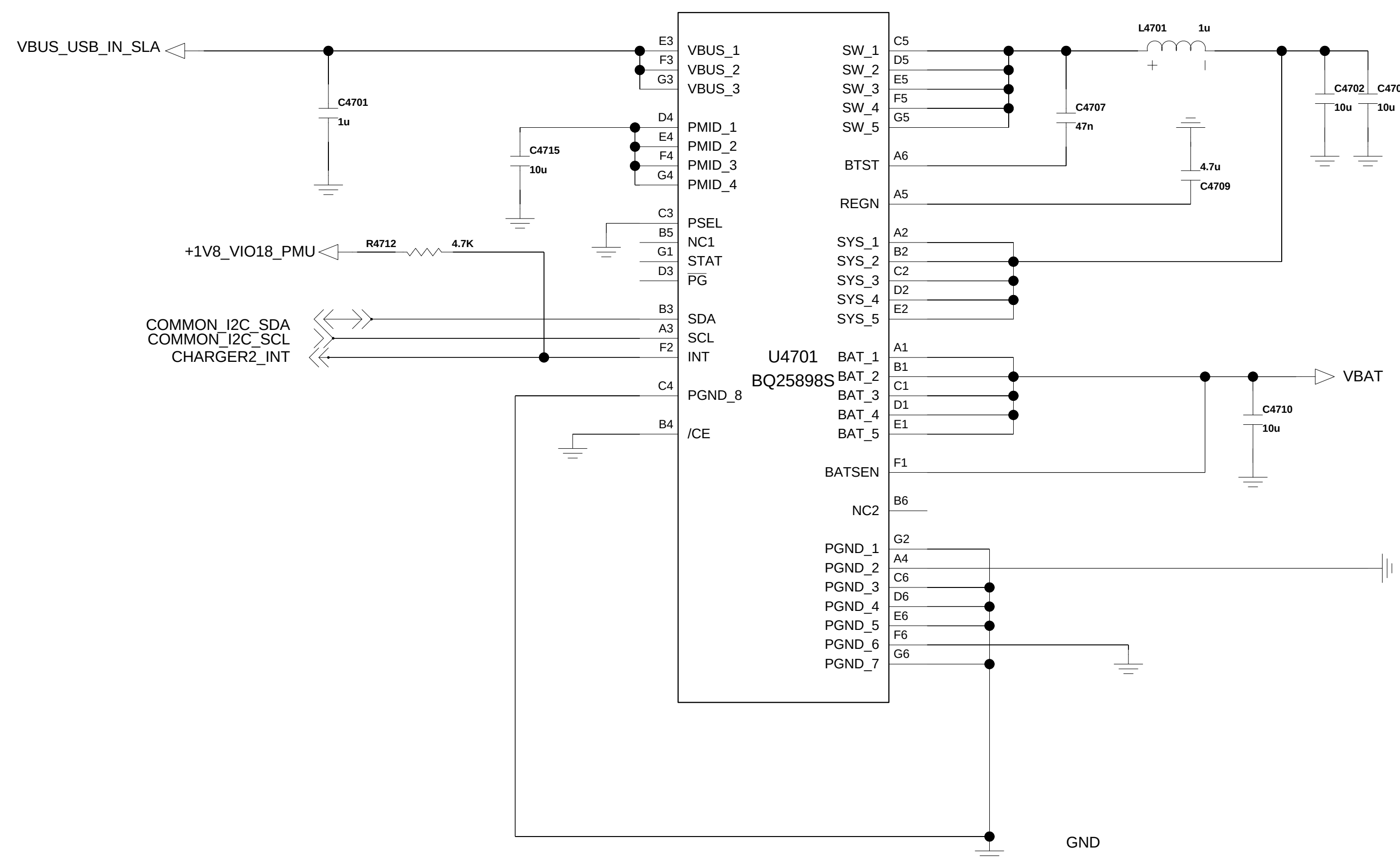
<4-1-15-1_PMIC_MT6353_POWER > Rev_0.1



< Parallel Charging - SM5424 >



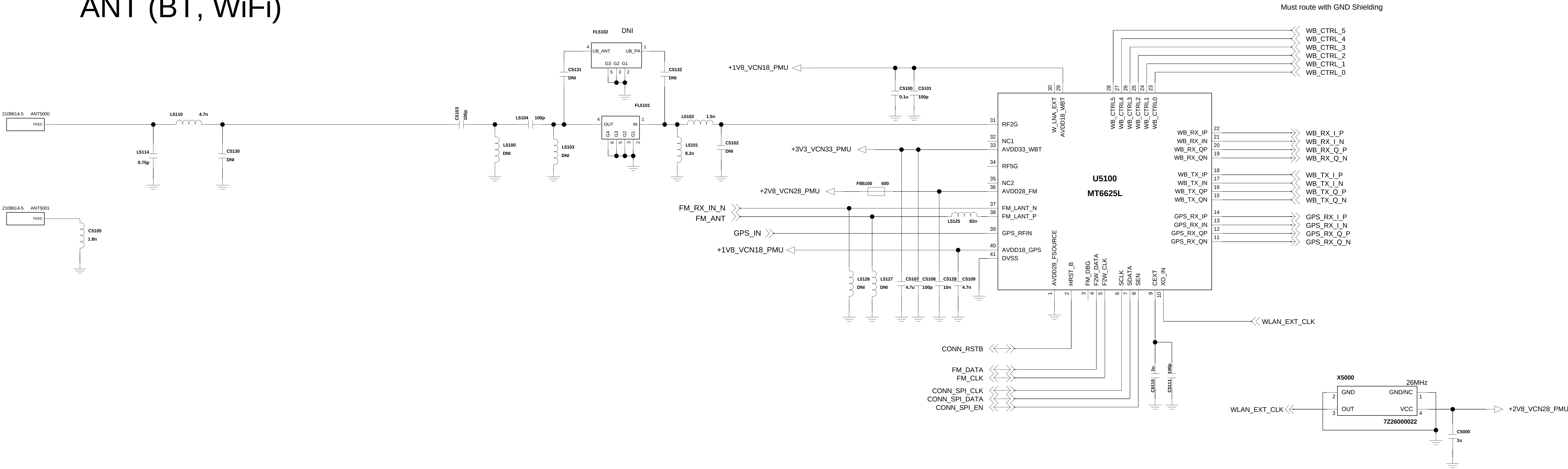
< SLAVE CHARGER bq25898s >



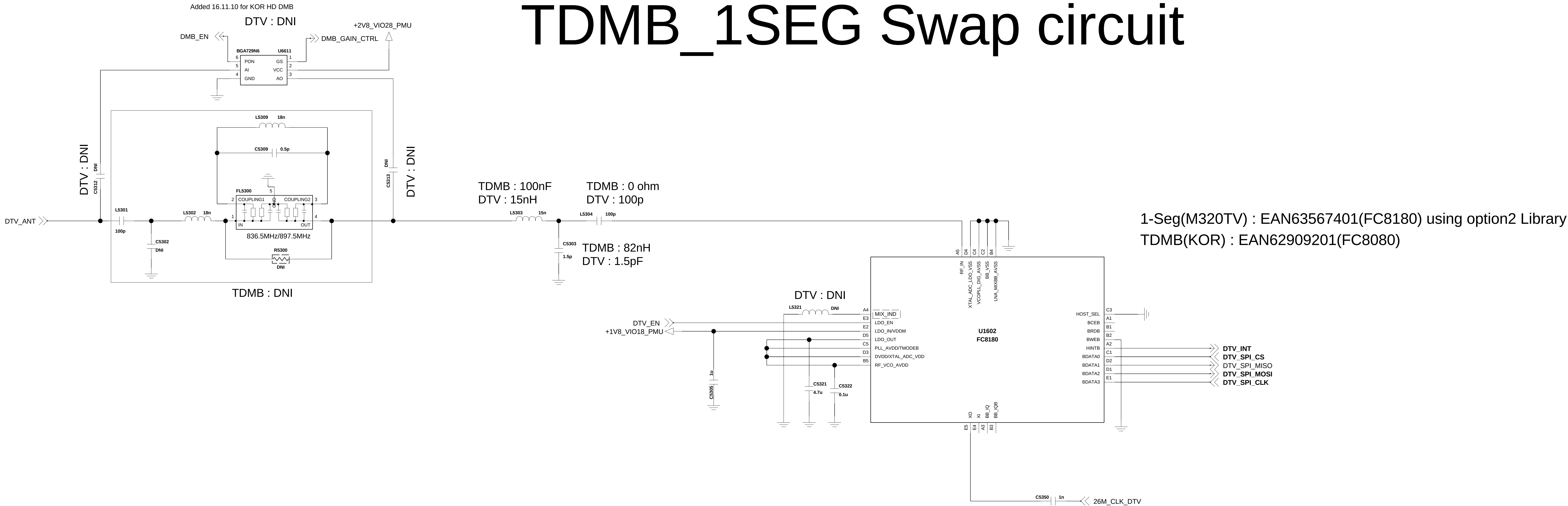
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Rev_0.3

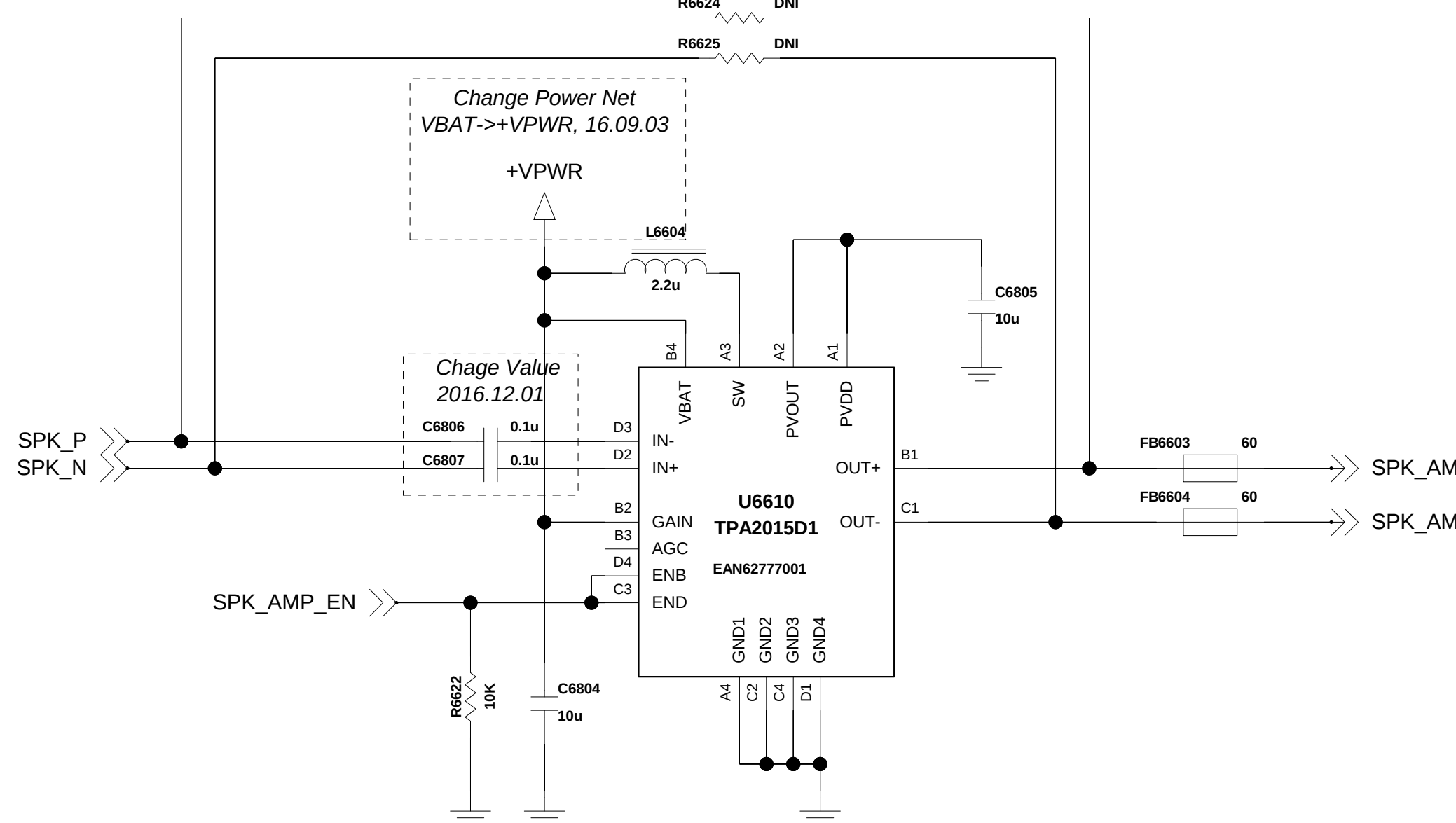
ANT (BT, WiFi)



TDMB_1SEG Swap circuit

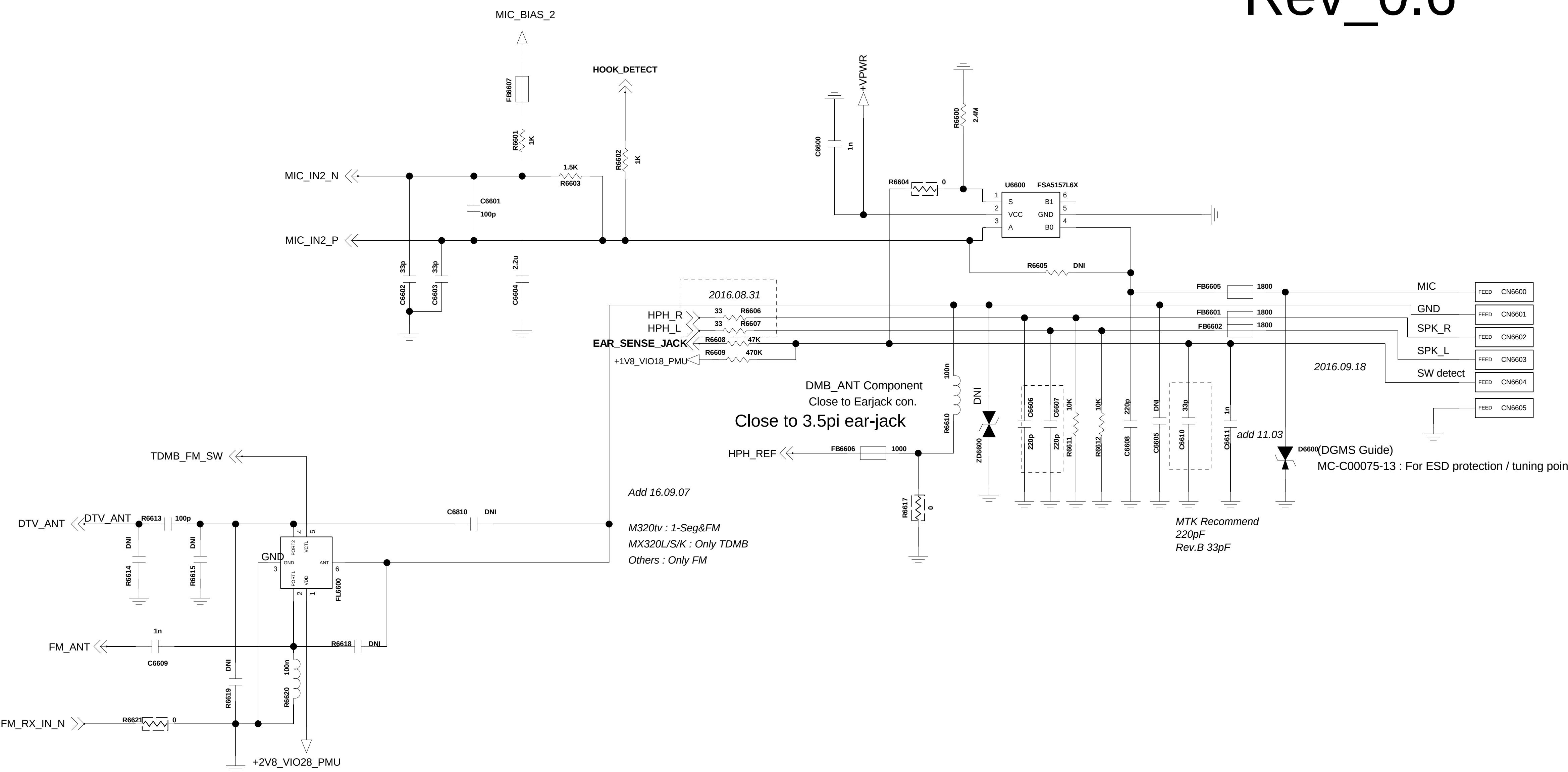


< SPK AMP >

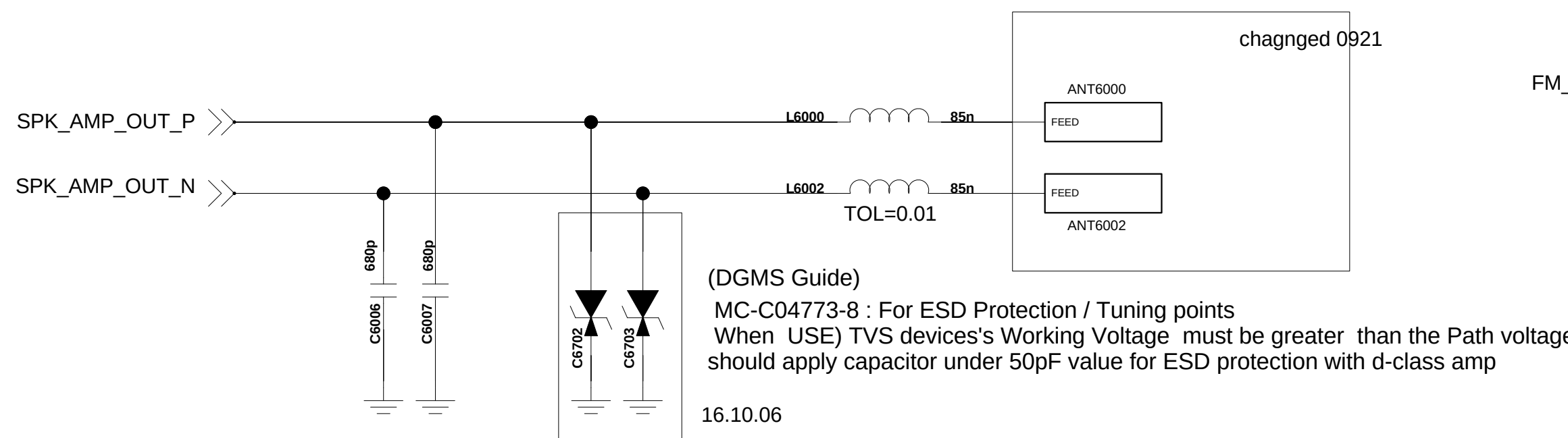


< 6-6-1_Earjack >

Rev_0.6



< 6-7-1_Speaker >



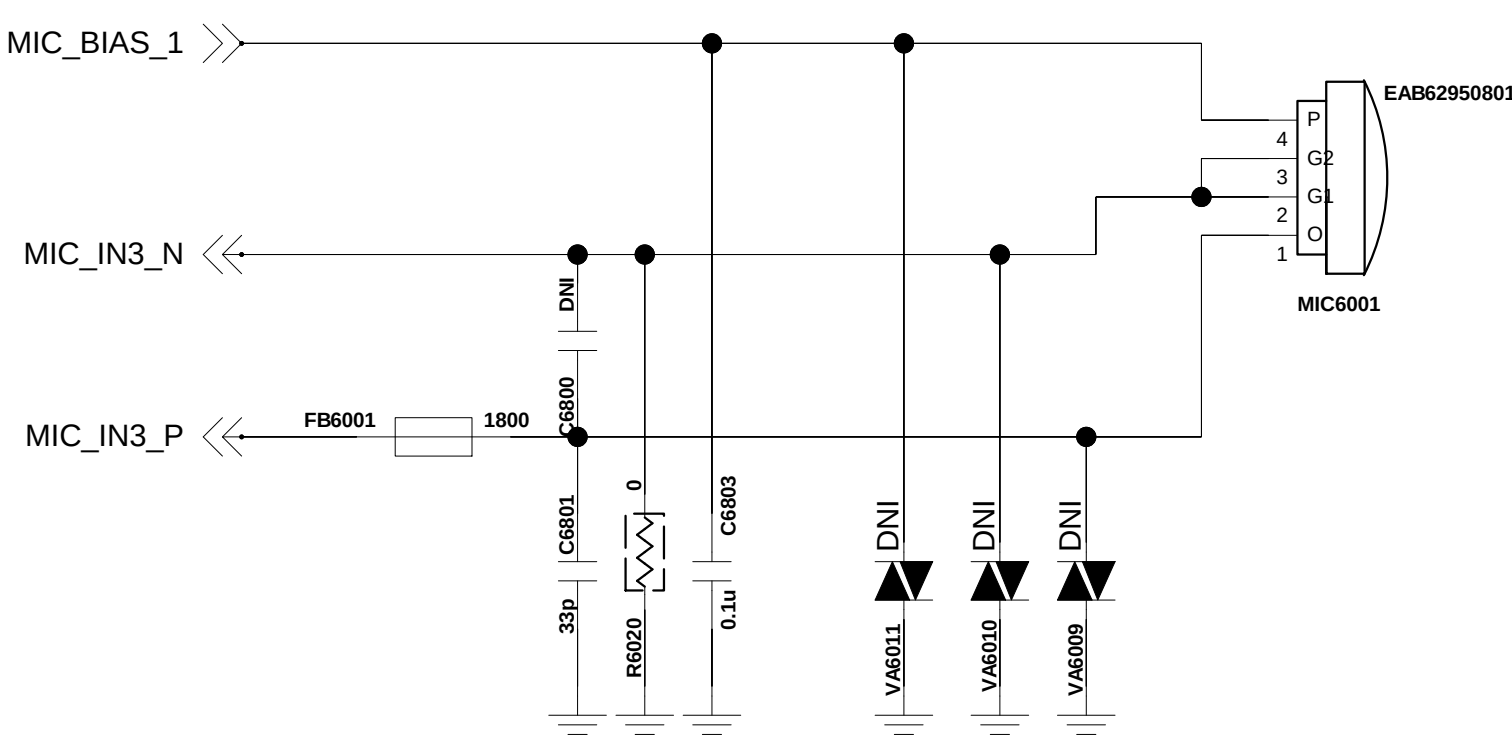
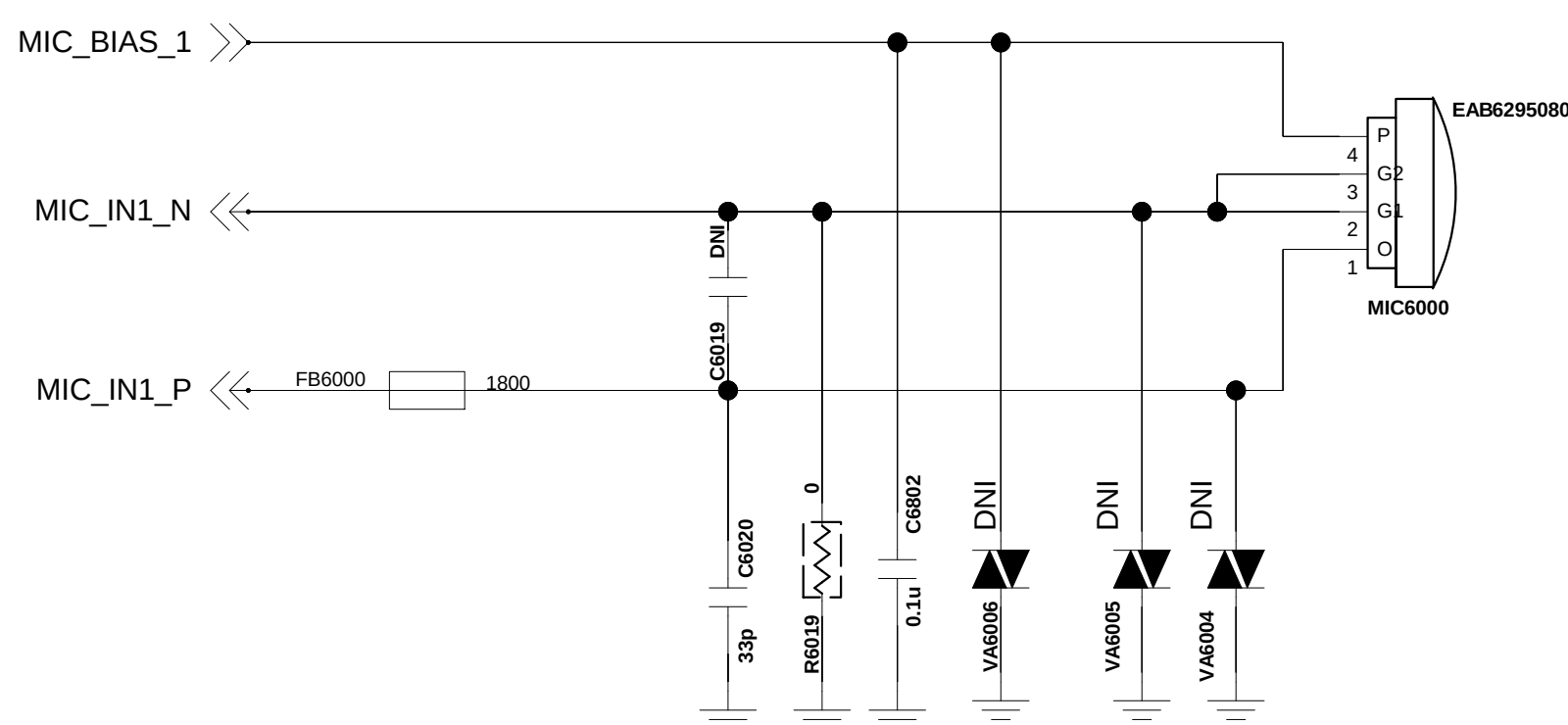
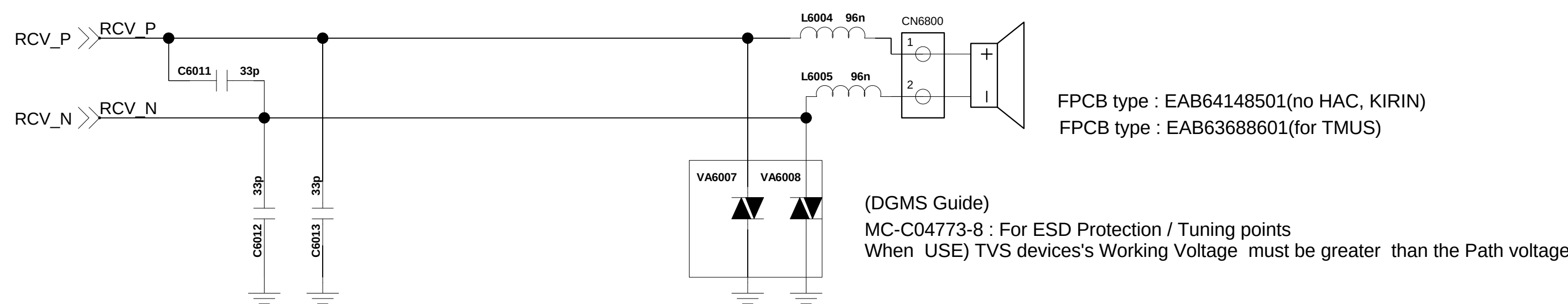
< 6-9-2_2MIC >

Main MIC

Sub_MIC

< 6-8-2_Receiver >

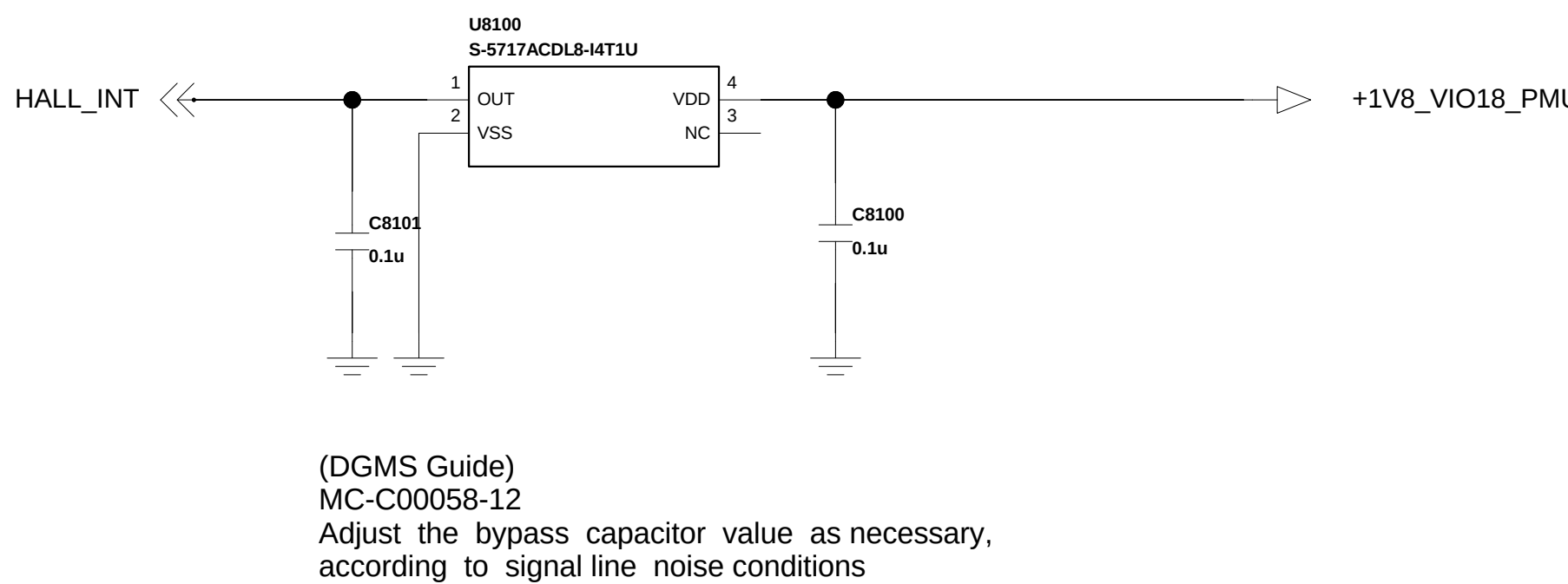
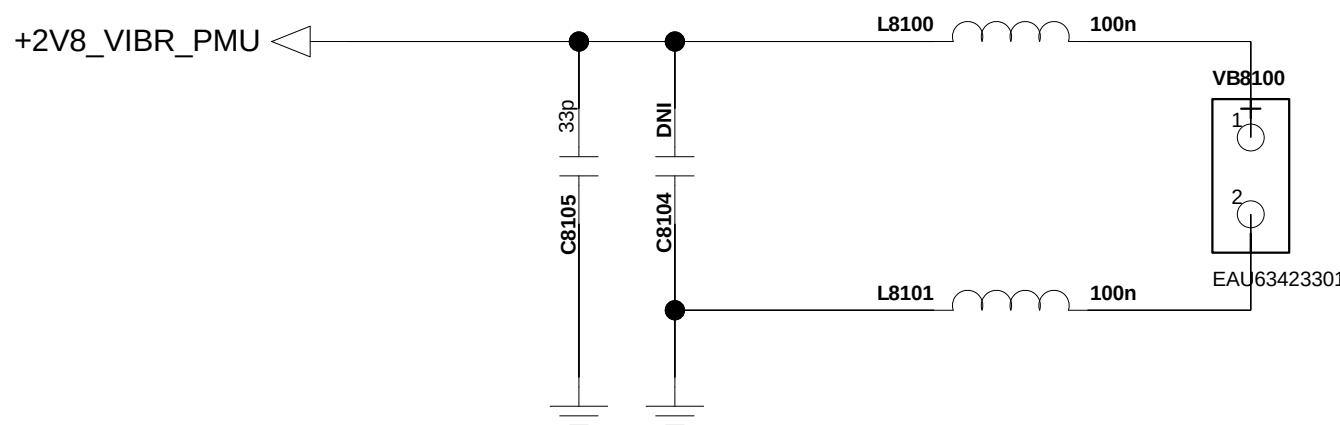
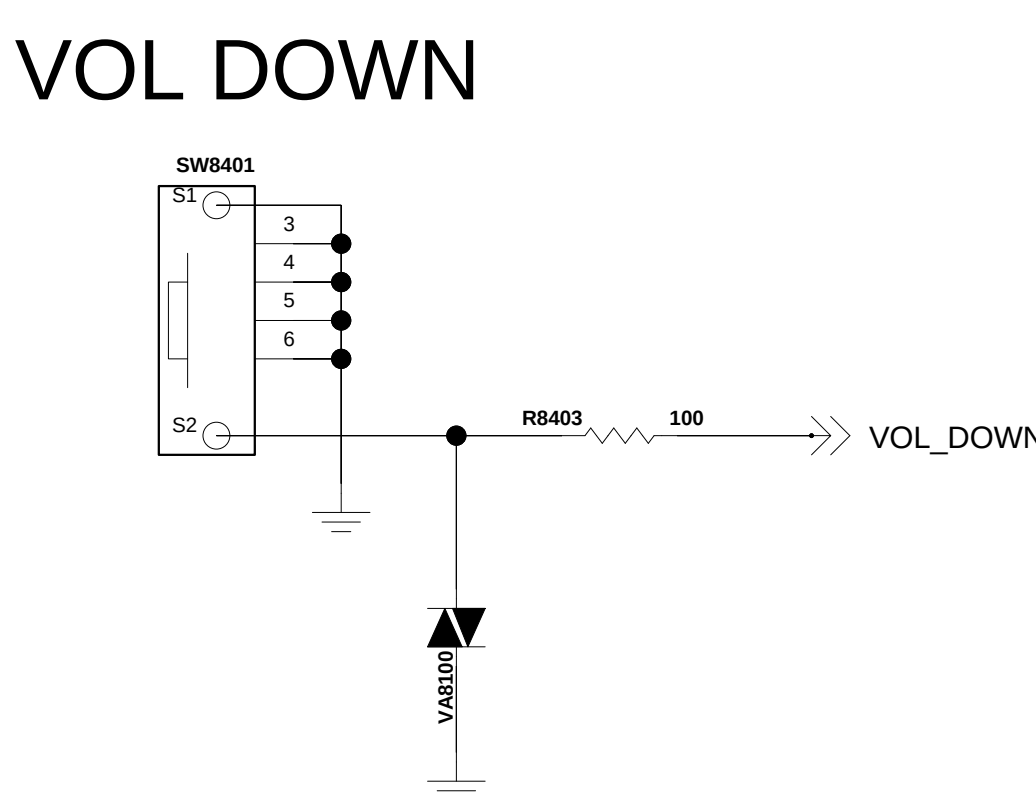
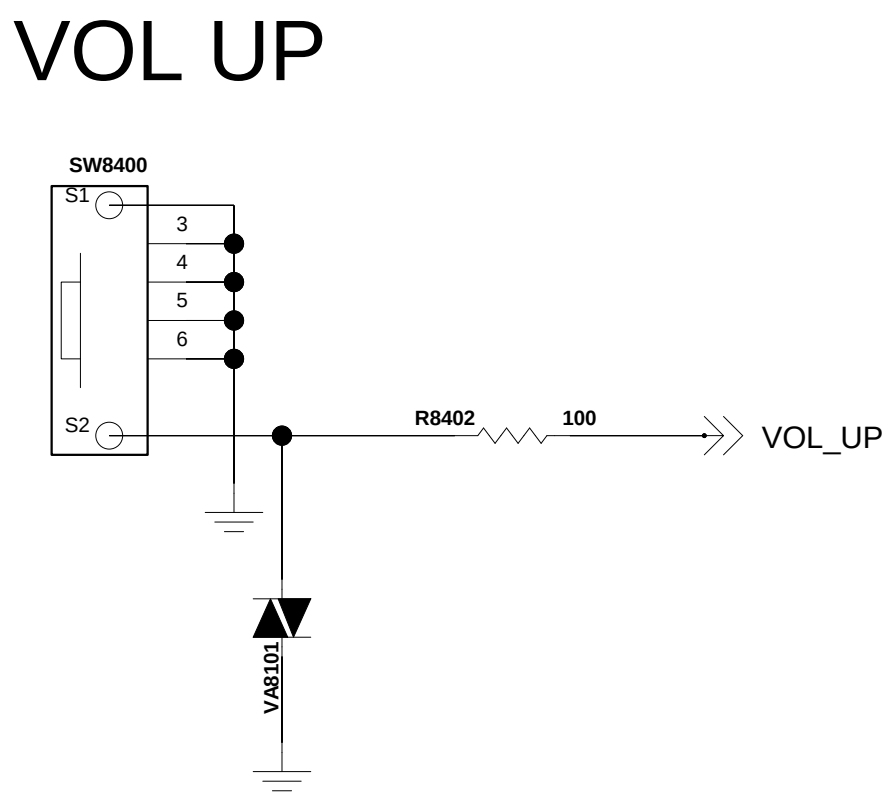
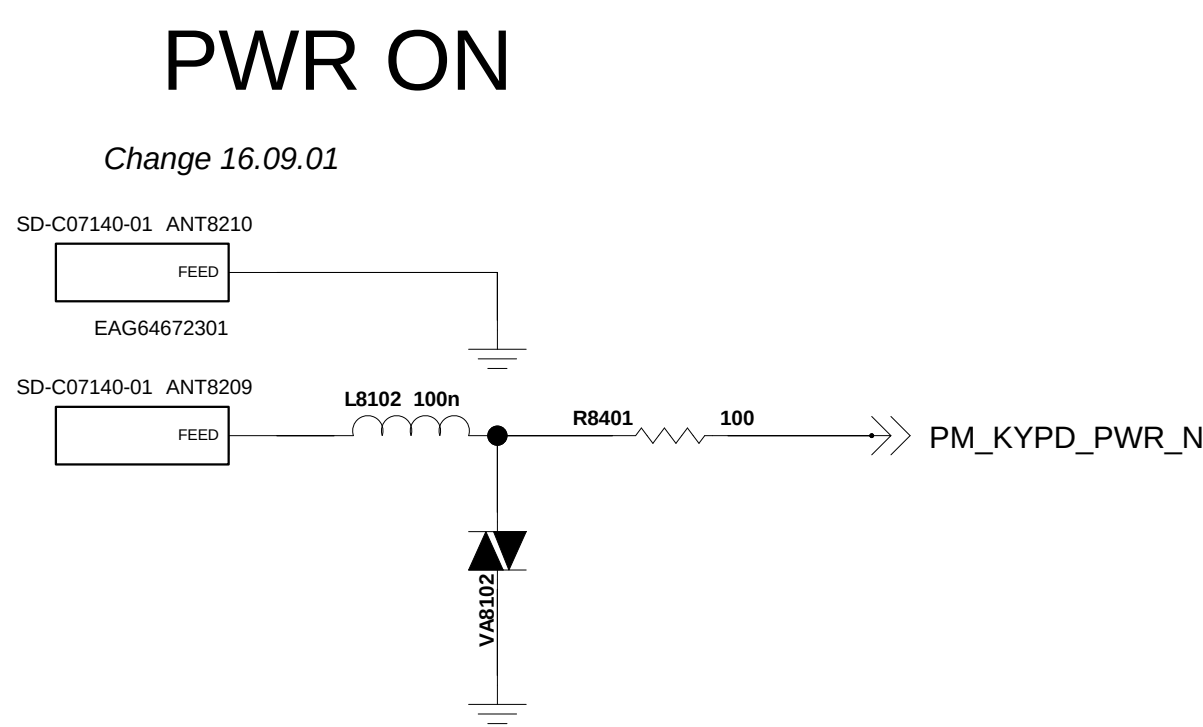
Rev_0.4



< PWR KEY_VOL KEY>

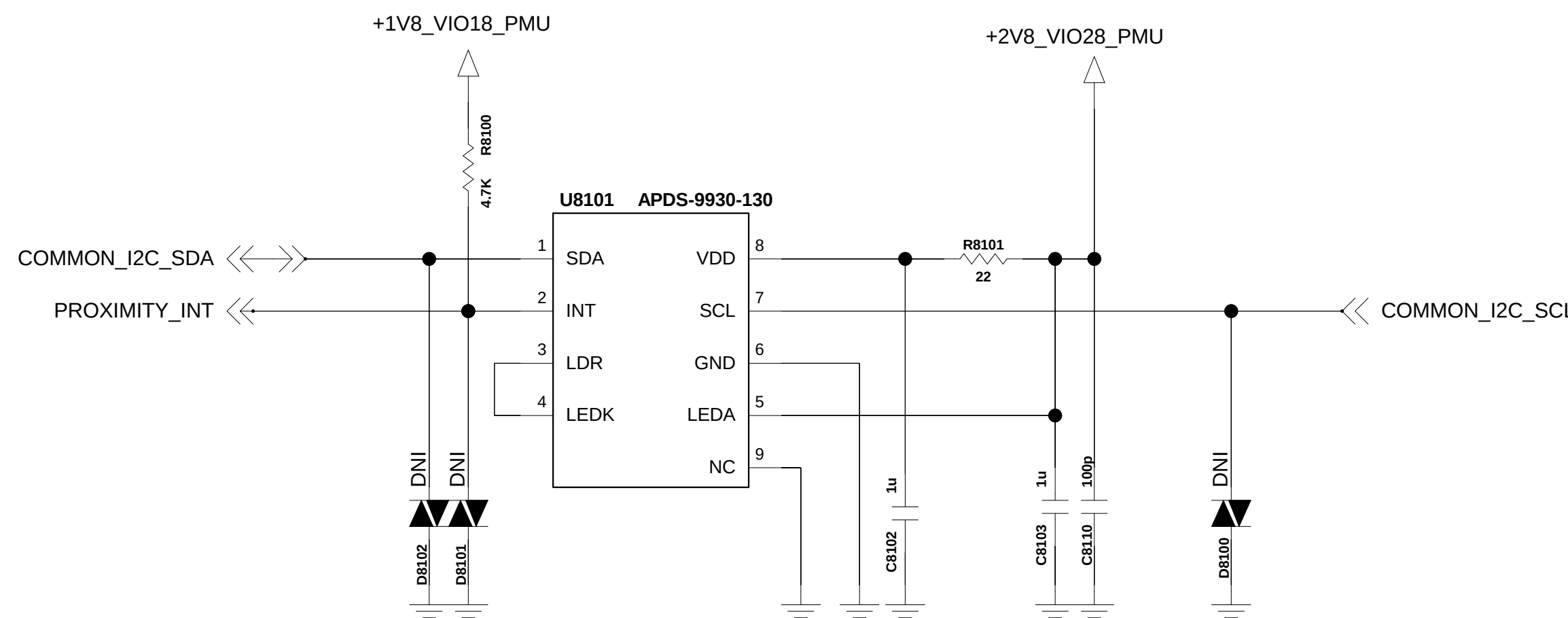
< 9-9-1-1_MOTOR >

< 8-7-1-5_Hall_IC_S5717 > Rev_1.1

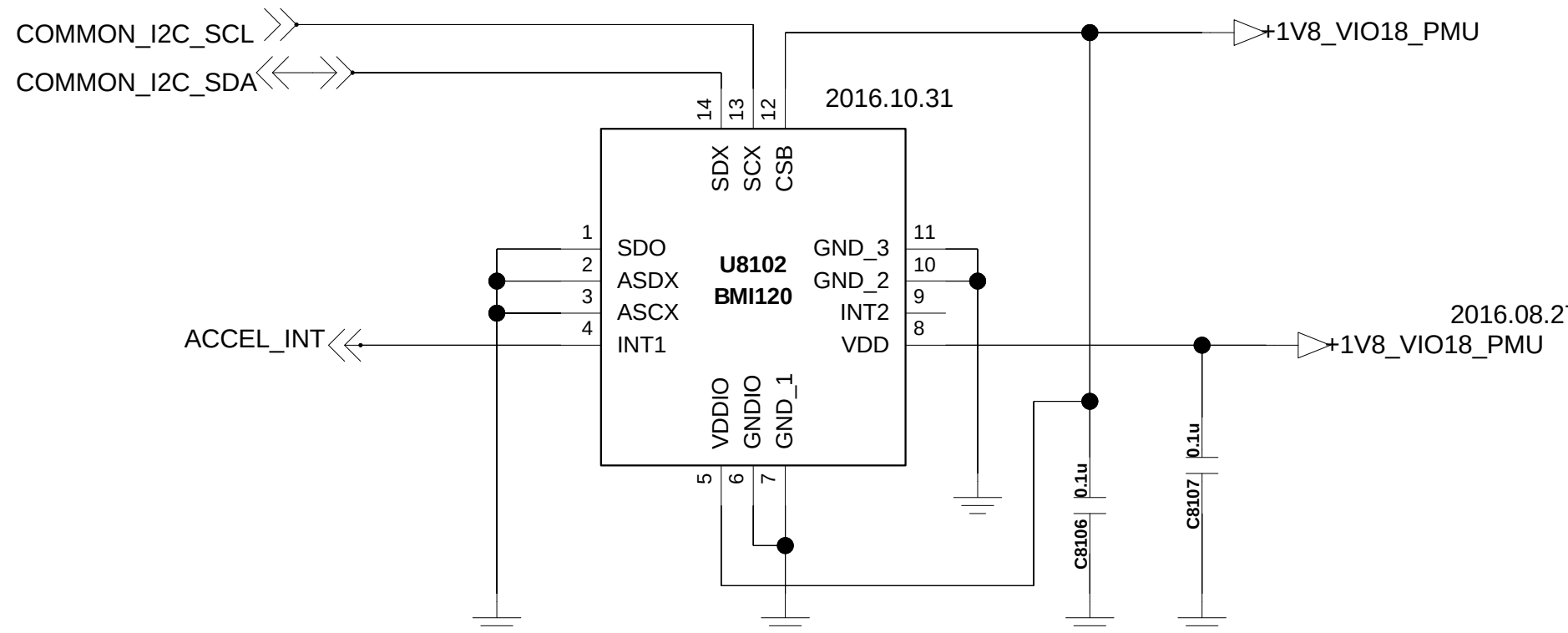


(DGMS Guide)
MC-C00058-12
Adjust the bypass capacitor value as necessary,
according to signal line noise conditions

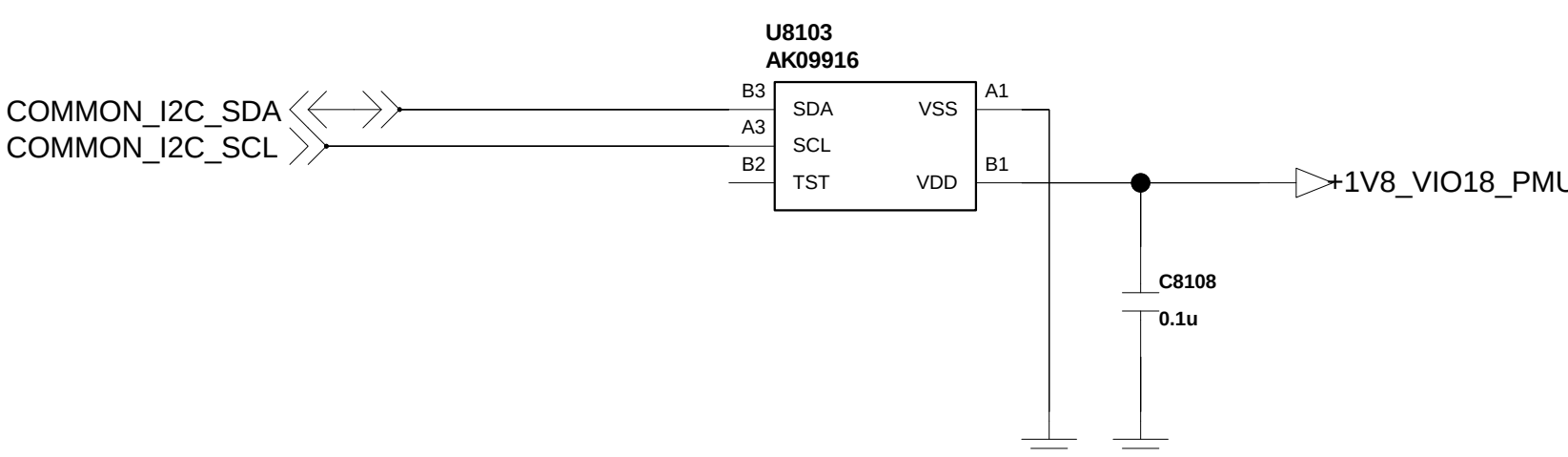
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<8-1-2-1_Accel_Gyro_BMI120>

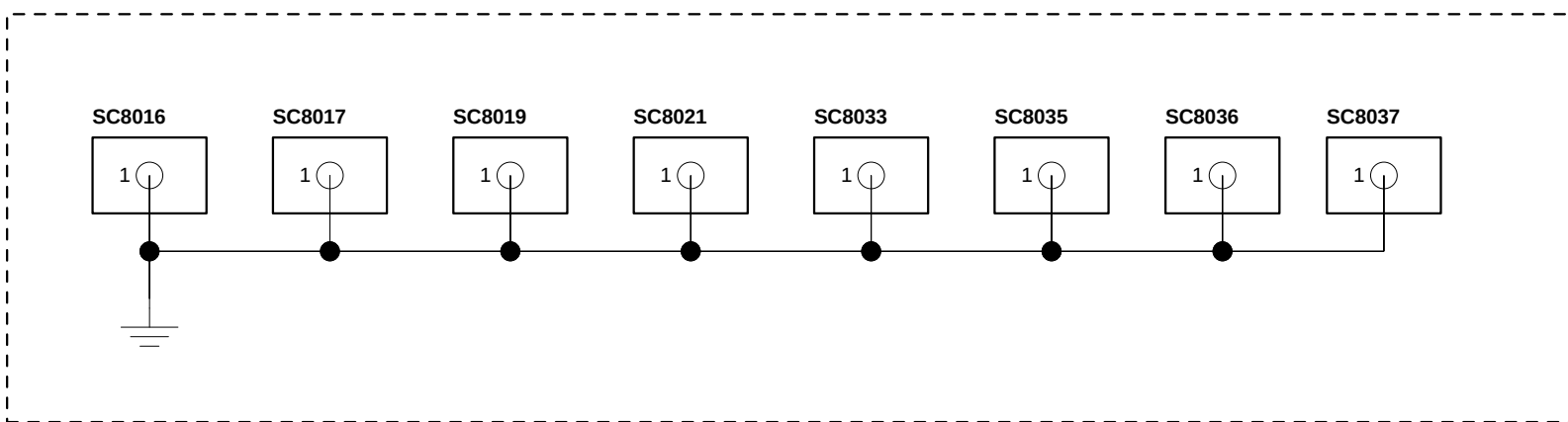


< Geomagnetic_AK09916C >



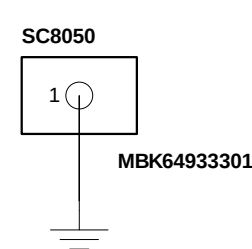
Shield CAN / C-Clip

MT6750/MCP Shield Can clip_8ea(BOT1)

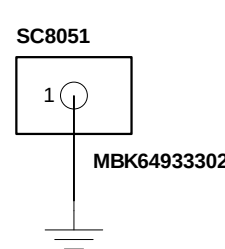


TOP SMT Shield Can

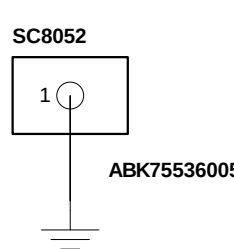
BT/WIFI SMT Shield Can(TOP1)



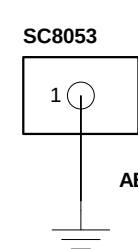
MIMO SMT Shield Can(TOP2)



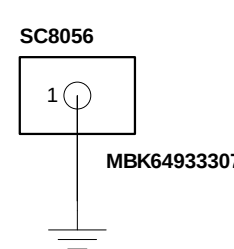
PMIC SMT Shield Can(TOP5)



1st Charger SMT Shield Can(TOP6)

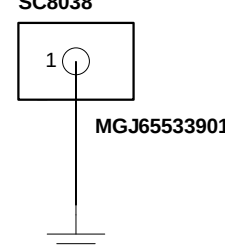


DTV SMT Shield Can(TOP7)

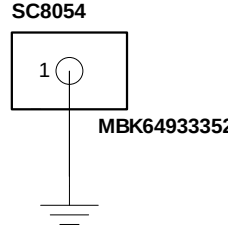


BOT SMT Shield Can

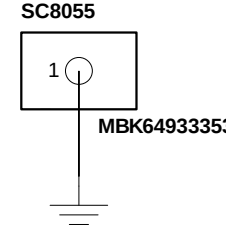
SUS Plate (beside BOT1)



MMPA SMT Shield Can(BOT2)

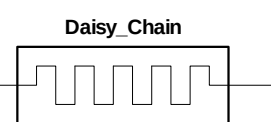


RF Block SMT Shield Can(BOT3)



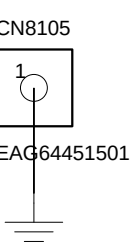
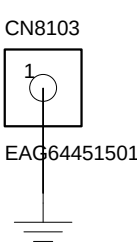
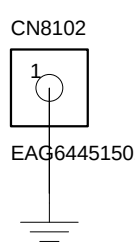
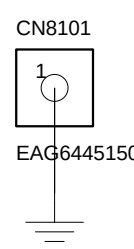
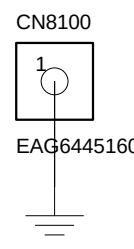
PID 6x6

DAISY_CHAIN



Add 16.09.03

SMT Gasket



For RSE

TOP Lower

Add 16.10.06

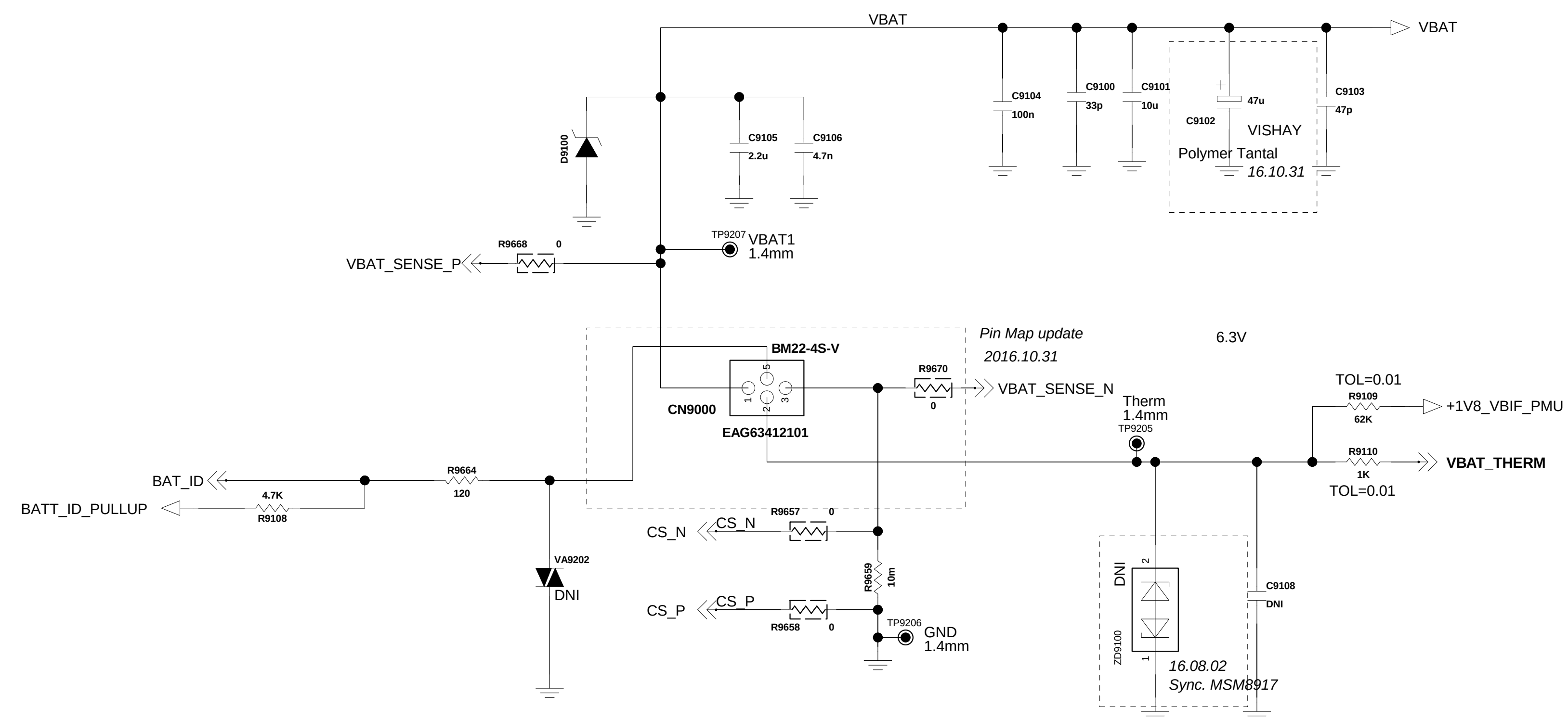
OJB009

OJB010

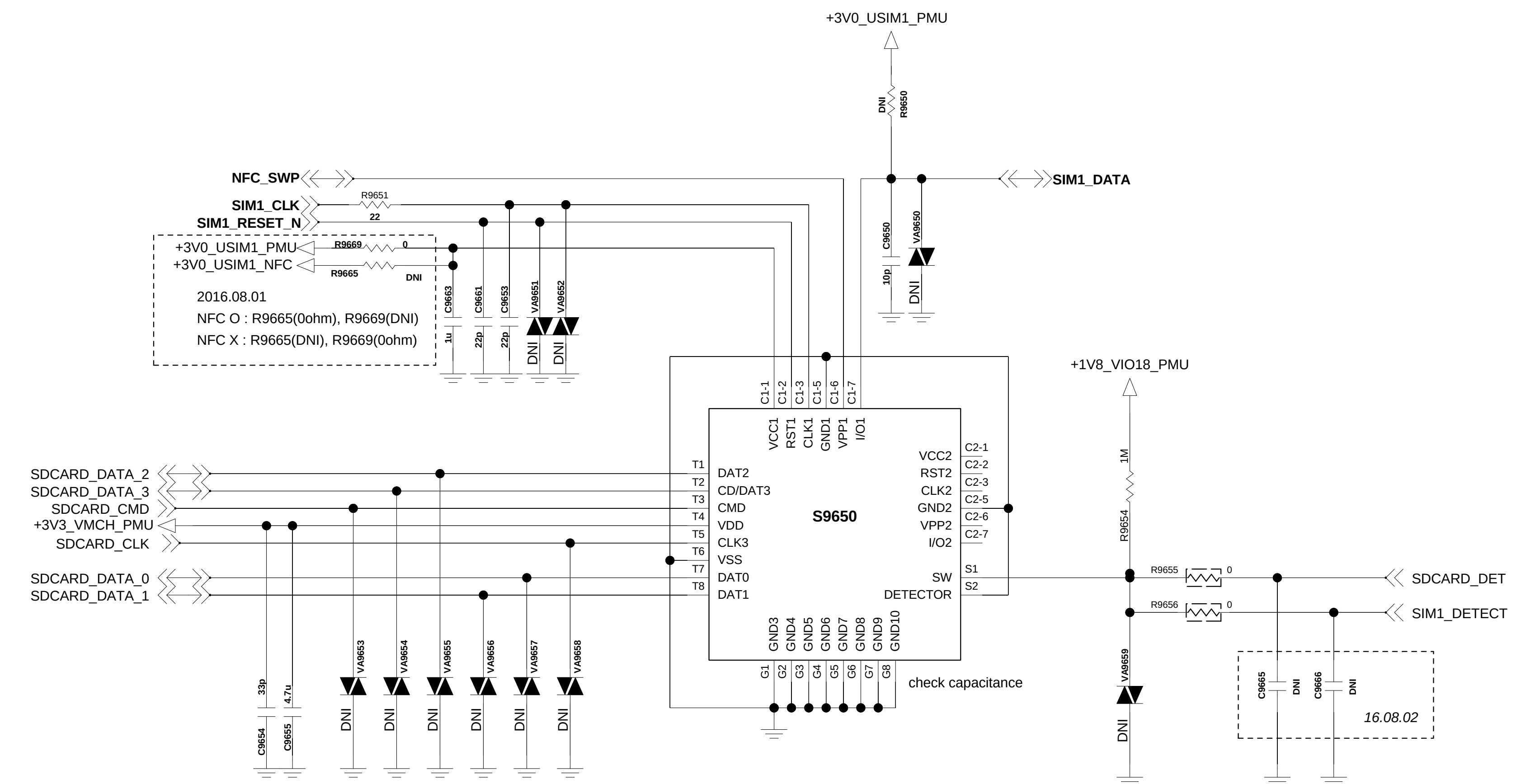
OJB011

OJB012

Circuit 4. Batt Conn. /w Embedded Batt



< 9-6_u_SDCARD_Socket > Rev.1.1
1st SIM & SD Card



(DGMS Guide)

MC-C04581-13 : Below items should be satisfied

1. SIM_DATA pull-up resistor
2. Use under 33pF (Cap_Varistor) on each signal line
3. When routing, SIM_DATA, SIM_CLK should be seperated
4. CAP,Varistor should Place close to connector

(DGMS Guide)

MC-C04582-14 : Below items should be satisfied

1. SD_DETECT Pull-up or Pull-down
2. Data/CMD line should be needed Pull-up resistor
(If chipset has internal Pull-up, comment it on circuit)
3. Use under total 10pF Capacitance on each line ; Data [0:3], CMD - Cap. TVS diode, varistor
(on other line, We recommend 10pF capacitance and use under 30pF capacitance)
4. CAP_Varistor should Place close to connector

(DGMS Guide)

MC-C05446-7 : Below items should be satisfied

1. Power of SD card should use independent Power source just like LDO/PMIC which can turn off it.
2. Per chip type which support SD card should manage current capacity.
(In case of High speed mode, SD Card Power should get margin of 200mA)
3. Do not connect serial component to SD card power, voltage drop can arise at Power line.

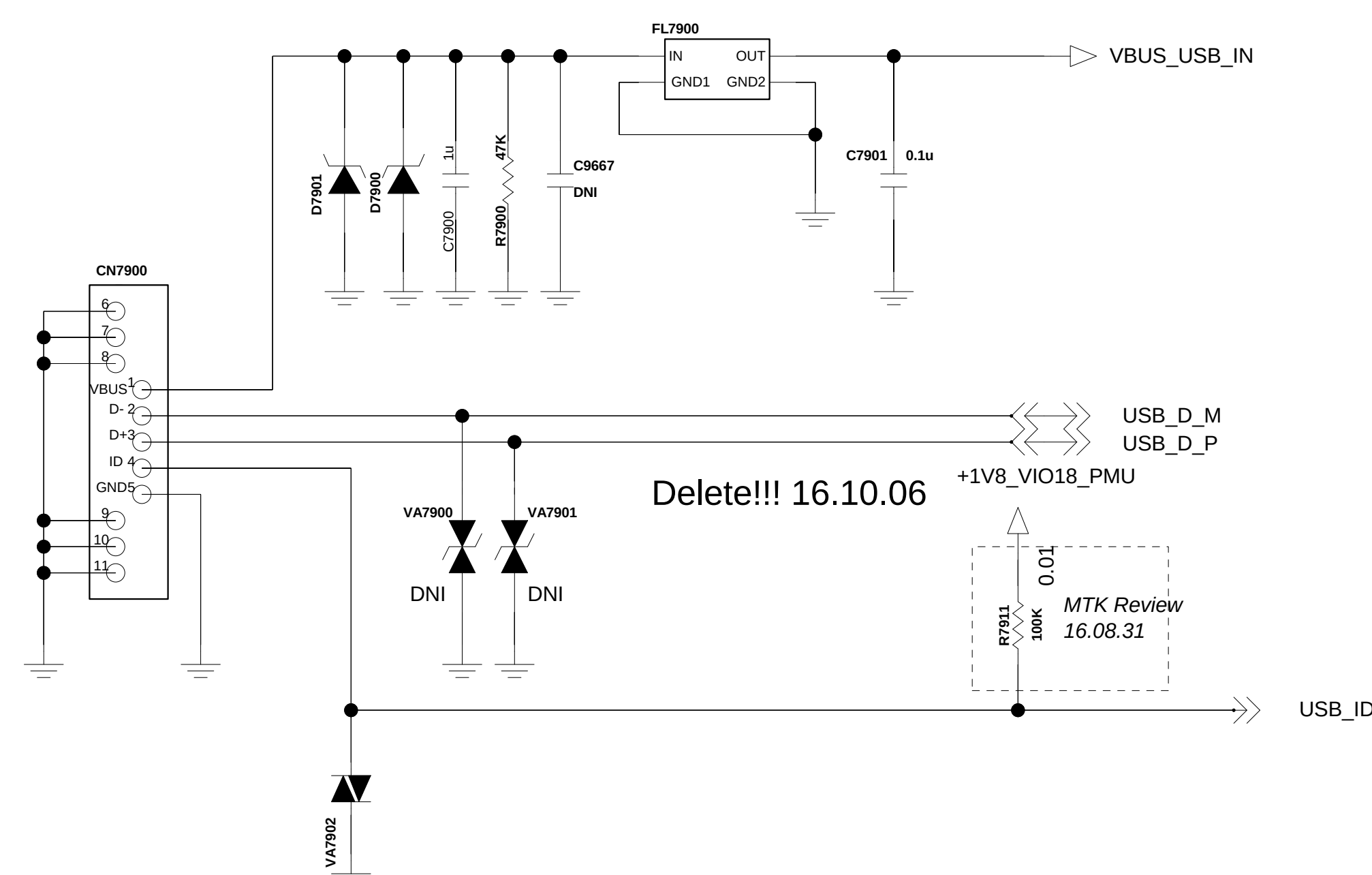
<9-2-1_USB_IO> Rev.1.0

Connector 1. USB 2.0

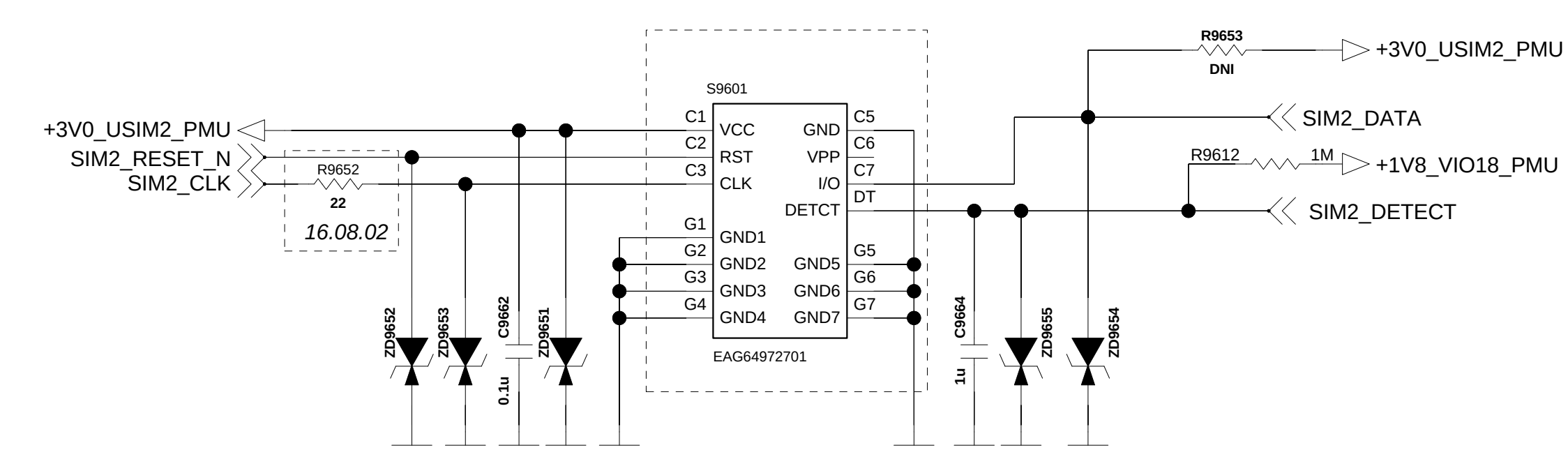
<4-3-3_OVP_RT9718> Rev.0.3

Delete!!! 16.09.03

FACTORY TP
Delete 16.08.30



NanoSIM Socket (2nd SIM)

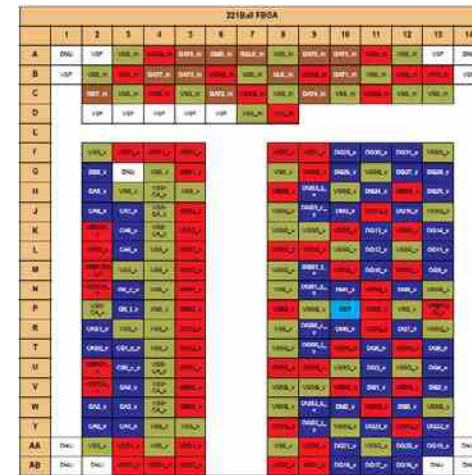


6. BGA PIN MAP

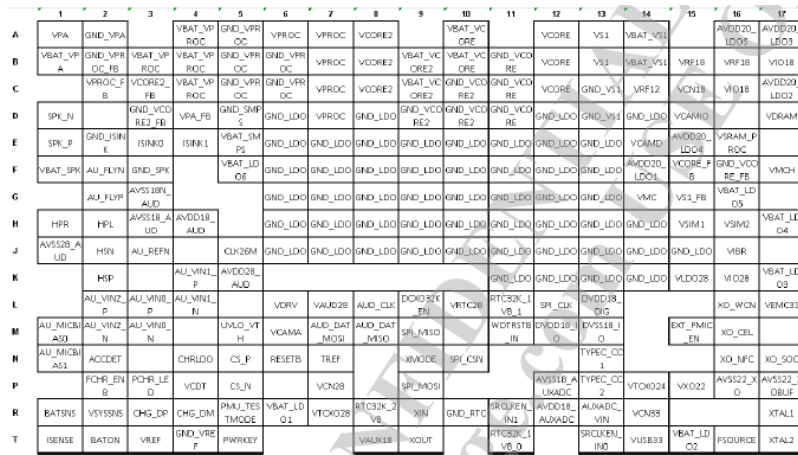
U2000_MT6750_IC,Digital Baseband Processor(Top View)



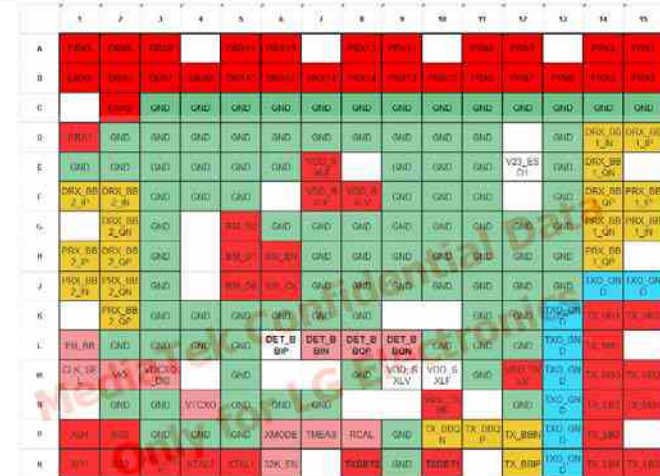
U3300_KMQX10013M-B419_IC, MCP EMMC (Top View)



U4100_MT6353_IC,PMIC (Top View)

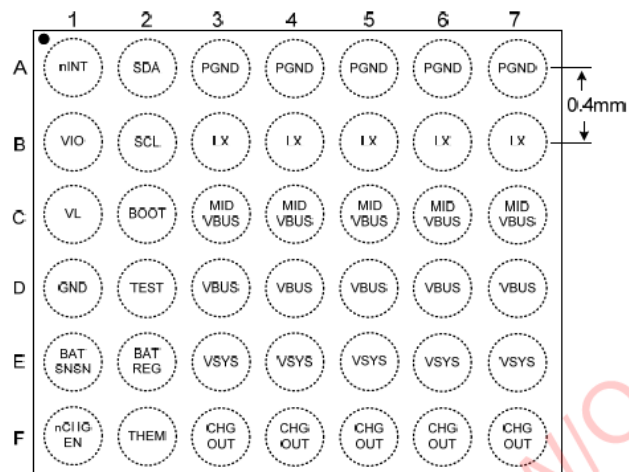


U15000_MT6176_IC,RF Transceiver,4G (Top View)



6. BGA PIN MAP

U4702_SM5424, IC Charger (Top View)



U6610_TPA2015D1, IC, Speaker AMP (Top View)

